

The Long Shadow of Seasonality: Climate Endowments, Agricultural Pathways, and Escape from the Malthusian Trap

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Abstract

How did pre-industrial economies absorb climate shocks? Population, land-use composition, and urbanization could each respond, yet existing work has tracked these margins separately. We build a new monthly paleo-economic panel for 196 countries reaching back to 1421 CE and use exogenous volcanic forcing 1500–1900 to identify a three-equation joint panel VAR for population, cropland-share, and urban-share growth. Cross-pathway asymmetry is the central finding. Each agricultural-development pathway adjusts along a different structural axis: traditional crop and pastoral systems absorb climate shocks demographically ($\hat{\beta} = -9.7 \cdot 10^{-5}$ per Tg of stratospheric sulfur injection on annualised log pop growth, $p < 10^{-9}$); high-density intensive systems rebalance cropland-versus-grazing composition without losing people; early extensifiers slow urbanization—the unified-growth-theory exit margin. The axis on which a society responds depends on where it already sits in the escape trajectory. Welfare-side evidence from an Allen real-wage panel 1421–1850, a preventive-positive-check decomposition that surfaces a sanitation-mediated modern-era reversal, and a cross-country long-shadow regression ($R^2 = 0.45$) corroborate the central pattern. The same climate dimension that selected which agricultural system a society adopted in the deep past now selects which margin it uses to absorb climate shocks. Seasonality, properly measured, casts a long shadow.

JEL Codes: N50, O13, O44, Q54, J11.

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1 Introduction

Most of human history was Malthusian. Productivity gains were absorbed by population growth; living standards drifted within a narrow band. Then, beginning in eighteenth-century northwestern Europe, the trap broke. Income per capita rose without compensating fertility, and the modern era of sustained growth began. Why did some economies escape while others stayed inside? Unified growth theory has converged on a short list of candidate answers—human capital accumulation (Galor and Weil, 2000), late marriage and urban disease in northwestern Europe (Voigtländer and Voth, 2013), gradual selection on individual industriousness (Clark, 2007), New-World-crop relaxation of the Malthusian ceiling (Nunn and Qian, 2011), and the institutional inheritances that translated productivity into welfare rather than headcount (Ashraf and Galor, 2013)—each of which has a

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climate sub-text but none of which has been tested against the actual climate that pre-industrial societies experienced.

This paper asks a sharper question. Pre-industrial economies could in principle absorb a climate shock along any of three structural margins: their population could fall, their land use could rebalance from intensive cropping toward pastoralism or vice versa, or their structural transformation out of agriculture could speed up or slow down. We ask which margin actually moved, and whether different economies moved different margins. The answer, identified off exogenous volcanic forcing 1500–1900 in a three-equation joint panel VAR, is the central finding of the paper: economies absorbed climate shocks along *different structural axes depending on where they already sat in the unified-growth-theory escape trajectory*. Traditional crop and pastoral systems took the hit on population. High-density intensive systems rebalanced land use without losing people. Early extensifiers—countries already partway out of agriculture—slowed urbanization. Each pathway picked its own margin, and the pick maps cleanly onto the pathway’s position in the escape trajectory.

The empirical exercise requires data that the literature has not previously been able to assemble. We build a new monthly paleo-economic panel that aggregates the ModE-RA paleo-reanalysis (Franke et al., 2023) onto HYDE 3.5 land-use cells (Klein Goldewijk et al., 2017), and link the resulting climate-and-land-use panel to the Sigl-Toohey eVolv2k volcanic record (Sigl and Toohey, 2024; Toohey and Sigl, 2017), the Brecke conflict catalogue (Brecke, 1999), and the Allen-Nuffield wage-and-price archive (Allen, 2001). The panel covers 196 countries and 3,187 sub-national units from 1421 to 2025, at monthly climate resolution and decadal demographic resolution; we use it to run the within-country Malthusian econometrics that the previous literature has had to leave to plausible storytelling. The intellectual frame is Galor’s; the climate mechanism extends Matranga (2024) and Ashraf and Galor (2011). The new contribution is the empirical apparatus and the three-margin asymmetry it surfaces.

The headline lives in Section 4. The supporting evidence is layered. Volcanic cooling reproduces the pathway-heterogeneous demographic response across 1500–1900 and survives extension back to the 1257 Samalas eruption using city-level wheat prices with region-aware exposure weighting and conflict-and-pandemic controls. An Allen real-wage panel for six European countries 1421–1850 returns the canonical Malthusian price-on-population coefficient at city resolution. A decomposition of the Malthusian response into preventive (fertility) and positive (mortality) checks on annual European data finds the response dominated by the positive check on infant and child mortality—the historically documented summer-mortality pattern—and that the coefficient collapses monotonically to zero by 1950–2022 as sanitation, refrigeration, and antibiotics develop, confirming the mechanism. Pre-industrial inter-annual climate volatility 1421–1750 then emerges as the single strongest cross-country predictor of modern population growth at $R^2 = 0.45$, surviving latitude and an extended deep-determinant battery. Seasonality, properly measured, casts a long shadow.

The paper proceeds as follows. Section 2 introduces the new paleo-economic panel and the five-pathway agricultural typology that organises the rest of the paper. Section 3 works out the within-country identification strategy and runs the storage-demand subnational regression that motivates the headline. Section 4 estimates the three-equation joint VAR, develops the cross-pathway asymmetry, and corroborates it with a discrete volcanic event study, the Allen real-wage panel, and a preventive-positive-check decomposition. Section 5 runs the long-shadow regression. Section 6 consolidates robustness; Section 7 concludes. The appendix collects the theoretical framework, the calibrated quantitative model and counterfactuals, the pre-XV-century volcanic extensions, the war-and-pandemic robustness, and a CamPop family-reconstitution annual series 1541–1850 that pushes the preventive-positive decomposition back into the deep pre-industrial regime.

2 A new monthly paleo-economic panel

The paper rests on a paleo-economic dataset that we assemble for the first time and that does most of the empirical work. We describe its construction, validate its alignment with the modern instrumental record where they overlap, and then use it to characterise the climate-pathway typology that organises the rest of the paper. The dataset itself is, we believe, the more durable contribution: it can support econometric questions about pre-industrial development that the existing climate-history literature has been unable to address.

2.1 Construction

The dataset has three substantive layers and three event layers, all merged onto a common country-year (or city-year, or sub-unit-year) grid. The first substantive layer is HYDE 3.5 (Klein Goldewijk et al., 2017), which provides gridded estimates of population, cropland, grazing land, irrigated land, rice cultivation, and urban population at 5-arcminute resolution from 10,000 BCE to 2025 CE under three uncertainty scenarios. We aggregate to country totals using the `iso_cr` country raster and to 3,187 sub-national units using the `sub_iso_cr` raster. The HYDE temporal grid is irregular: 100-year intervals before 1700, decadal 1700–1950, and annual thereafter. The pre-industrial decadal sample is what powers most of the regressions; we use the century-spaced pre-1500 sample only in the appendix extension.

The second layer is ModeE-RA (Franke et al., 2023), a monthly global paleo-reanalysis covering 1421–2008 CE that assimilates documentary records, tree rings, ice cores, corals, and historical instruments into a coupled numerical-model background, returning monthly temperature and precipitation anomalies relative to the 1901–2000 mean on a 192×96 Gaussian grid. We aggregate each ModeE-RA cell onto HYDE country and sub-national units via cosine-latitude-weighted area sums on the 5-arcminute country raster; the CRU TS 4.09 (Harris et al., 2020) 1901–1950 country-month climatology, computed analogously, is added back to the anomalies to obtain absolute monthly temperatures and precipitation throughout 1421–2008. We rebuild ERA5 (Hersbach et al., 2020) at the country level by area-weighting the raw 0.25° ERA5 monthly fields onto HYDE country boundaries; per-country mean bias is removed by a constant offset estimated in the 1950–2008 overlap with ModeE-RA.

Three derived climate measures play headline roles in what follows. The intra-annual temperature range $\sigma_s^T(i, t) = \max_m T_{i,t,m} - \min_m T_{i,t,m}$ captures within-year variation. The productive-months index $\text{Prod}_{i,t}$ counts calendar months in which $5^\circ\text{C} \leq T_{i,t,m} \leq 30^\circ\text{C}$ and $P_{i,t,m} \geq 30$ mm—months simultaneously warm enough and wet enough for rain-fed cropping; the complement $12 - \text{Prod}_{i,t}$ is the non-productive-months count and captures storage demand. The inter-annual temperature volatility $\sigma_v^T(i, t)$ is the country-specific standard deviation of annual mean temperature in a 30-year centred window. In within-interval panel regressions we use within-interval volatility rather than the rolling window. We report a published-criterion alternative based on FAO-style wheat suitability in Section 6.

The three event layers attach to specific (country, year) or (city, year) cells of the panel. The Sigl-Toohey eVolv2k v4 ice-core record (Sigl and Toohey, 2024; Toohey and Sigl, 2017) provides annual estimates of global stratospheric volcanic sulfur injection from 500 BCE to 1900 CE, with eruption latitudes for each event; we link this to the climate panel as both a discrete event-study indicator and a continuous distributed-lag regressor. The Brecke Conflict Catalogue (Brecke, 1999) adds 3,708 violent conflicts 1400–1999 with start- and end-years and free-text actor descriptions, plus 1,147 European conflicts 900–1402 in a supplementary extension. We string-match the actor descriptions to nine medieval regions to produce binary war indicators per (city, year), and hand-code 30 sieges

of specific panel cities and 49 plague episodes from standard catalogues. The Allen-Nuffield wage-and-price archive (Allen, 2001) contributes silver-content-normalised wheat prices for 17 European cities 1259–1914 and real-wage series for 8 cities 1264–1850; we parse the silver-grams-per-hectoliter wheat-price column directly from each city’s xls file and augment with Tuscan prices from Malanima (2002) and the Roman-Egypt wheat-price series of Harper (2016).

2.2 Validation

The country-level monthly climate panel matches the modern instrumental record where they overlap. In the 1950–2008 ModeE-RA/ERA5 overlap (Figure 1), the median country-month mean bias is 0.61°C , the median country-monthly T correlation is 0.971 (77.6% of countries exceed $r = 0.90$), and the median country-annual anomaly correlation is 0.938 after detrending. The agreement is strong enough that we treat ModeE-RA + CRU climatology as a substantive measurement of country-monthly climate rather than as a stand-in. ModeE-RA’s ensemble 1σ uncertainty in country-month temperature, propagated through the area-weighted aggregation, is on the order of 0.9°C in 1421–1500 and falls steadily to 0.37°C by 1850–1949; ModeE-RA precipitation uncertainty traces the same secular decline from 31.6 to 15.4 mm/month. Past 1950, we splice in our country-level ERA5 rebuild, which the bias-correction step described above brings onto the same scale as the paleo-reanalysis. Figure 21 reports the merged-panel uncertainty across 1421–2025, including the ERA5–ModeE-RA monthly RMSE in the 1950–2008 overlap as a cross-validation diagnostic. Section 6 discusses the implications for the headline regressions.

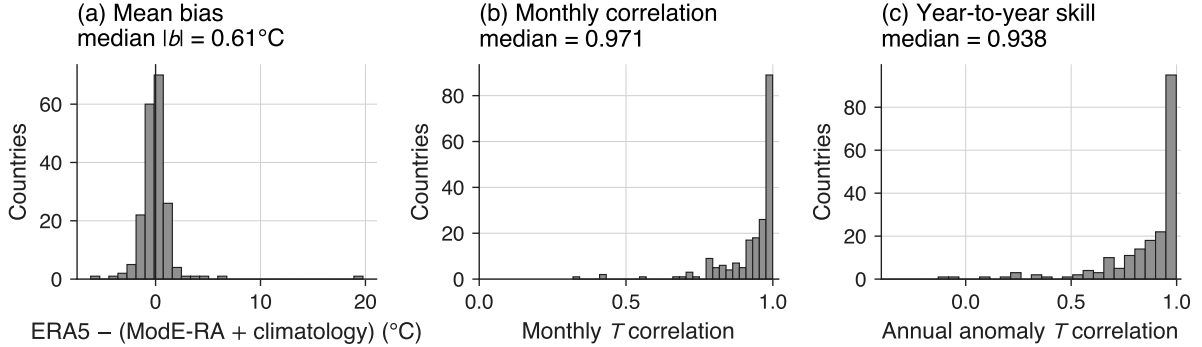


Figure 1: ModeE-RA versus country-level ERA5, 1950–2008. (a) Per-country mean bias (ERA5 minus ModeE-RA + CRU climatology). (b) Country-month temperature correlation. (c) Year-to-year anomaly correlation after country-wise detrending. Both panels use the same 196-country sample.

2.3 Five agricultural pathways

The new panel discriminates a typology of agricultural development that the existing literature has discussed informally but not measured systematically. Applying K -means clustering to standardised HYDE trajectory features—peak agricultural expansion year, density at 1750, 1000 CE, and 0 CE, crop share at 1750, rice share at 1750, irrigation share at 1750, urban share at 1750, two interval population-growth rates, and an agricultural-intensity change index—returns a five-cluster partition. The choice of $K = 5$ is the silhouette local maximum on this feature set (0.40 at $K = 5$, with a sharp drop to 0.23 at $K = 6$); bootstrap stability of the partition is 0.84 in median country-pair co-clustering probability across 200 resamples. The five pathways are high-density intensive (Japan, India, France, Germany, Netherlands, 16 countries), early extensifiers (Britain, Switzerland,

Austria, Denmark, Israel, 29 countries), crop-dominant late (Spain, Poland, Portugal, Thailand, 46 countries), pastoral/mixed late (China, Nigeria, Brazil, Iran, Mexico, 65 countries), and the irrigation pioneer Egypt as a singleton (Table 1). The pathways differ along all five clustering dimensions—peak agricultural expansion year, density at 1750, crop share, irrigation share, and rate of demographic expansion—in ways that match the historical literature’s informal taxonomy.

Table 1: Five agricultural transition pathways (HYDE 3.5 clustering)

Pathway	N	Density ₁₇₅₀	Crop share ₁₇₅₀	Peak ag. expansion	Representative
Crop-dominant late	46	11.8	80.6%	~1713	ESP, POL, PRT, THA
Pastoral/mixed late	65	6.6	22.0%	~1694	CHN, NGA, BRA, IRN
High-density intensive	16	62.8	58.8%	~1567	JPN, IND, FRA, DEU
Early extensifiers	29	14.1	48.9%	~ 783	GBR, CHE, AUT, DNK
Irrigation pioneer	1	4.4	100%	~1600	EGY

2.4 Climate signatures of the pathways

The new panel allows the first systematic measurement of pathway-level climate signatures. The natural Matrangla measure—raw intra-annual temperature range σ_s^T averaged over 1421–1750—does *not* significantly distinguish pathways at the country level. A one-way ANOVA on 153 countries (dropping the singleton Egypt cluster) yields $F = 0.50$, $p = 0.68$; a multinomial logit on σ_s^T , mean temperature, and growing degree days returns pseudo- $R^2 = 0.025$ and LLR $p = 0.39$. What does distinguish pathways is the joint structure of temperature and precipitation that determines storage demand. The productive-months index defined above differs sharply across pathways (Figure 2, Table 2):

Table 2: Climatically productive months by pathway, 1421–1750 (ModE-RA + CRU climatology)

Pathway	N	Productive months	Dry months	σ_s^T (°C)	σ_s^P (mm)
Crop-dominant late	45	9.54	2.46	10.5	196
Pastoral/mixed late	65	6.95	5.05	11.9	159
High-density intensive	16	9.10	2.90	13.0	175
Early extensifiers	27	7.67	4.33	12.8	152
Irrigation pioneer (EGY)	1	0.00	12.0	16.8	6

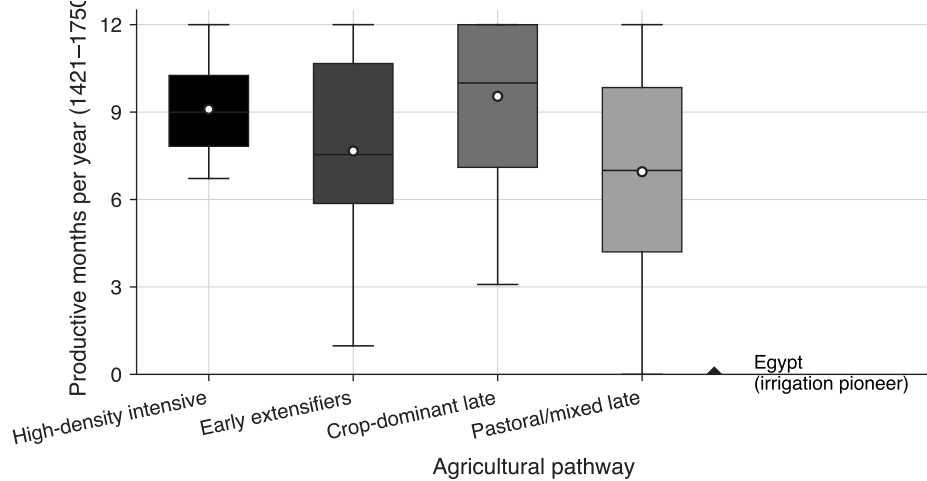


Figure 2: Number of climatically productive months per year (1421–1750) by pathway. The pastoral pathway concentrates in regions with few productive months; intensive crop systems in regions with many; Egypt, the lone irrigation pioneer, has none under rain-fed criteria. ANOVA $F = 6.31$, $p < 0.001$.

A one-way ANOVA on productive months returns $F = 6.31$, $p < 0.001$. A multinomial logit with the productive-months index and the storage index $((12 - \text{Prod}) \cdot \sigma_s^T / 12)$ attains pseudo- $R^2 = 0.065$ and LLR $p = 0.0003$. Productive months reduce the probability of pastoral selection ($\beta = -0.56$, $p < 10^{-3}$ relative to the crop-dominant baseline) and storage demand reduces the probability of pastoral selection ($\beta = -0.18$, $p = 0.014$). The lone irrigation pioneer (Egypt) sits at the extreme: zero rain-fed productive months but enormous storage demand—consistent with the historical record of state-led Nile irrigation as the institutional response to a climate that could not support rain-fed agriculture.

The Matranga story survives, in other words, but it operates through the *joint* structure of temperature and precipitation rather than through temperature variability alone. Storage demand is not a feature of cold winters by themselves but of the combination of growing-season productivity and the off-season’s harshness.

3 From storage demand to the Malthusian regression

The empirical work that follows is organised around the within-country identification strategy that the appendix model motivates. Appendix A sketches a two-stage model in the spirit of Ashraf and Galor (2011) that delivers two empirical predictions. First, in a society’s far past, the share of climatically productive months Π jointly with inter-annual climate volatility σ_v selects which agricultural system $\tau \in \{\text{intensive crop, extensive pastoral, irrigation-based}\}$ the society adopts; the model produces a productive-months threshold $\Pi^*(\sigma_v)$ above which intensive crop dominates pastoral (Appendix A, Equation 8). Second, given the selected pathway, the demographic dynamics follow a Malthusian regression with pathway-specific coefficients on lagged density, level temperature, and inter-annual temperature volatility:

$$\Delta \ln P_{it} = \alpha_i + \beta(\tau^*) d_{i,t-1} + \gamma(\tau^*) T_{it} + \delta(\tau^*) \sigma_v^T(T_{it}) + \eta(\tau^*) h_{it} + \varepsilon_{it}. \quad (1)$$

This is the regression we run. The Malthusian density coefficient $\beta(\tau^*)$ is predicted stronger in pastoral than in intensive systems—pastoral systems sit closer to a tight ceiling because they cannot

store, intensive systems carry capital that smooths the feedback—and the volatility coefficient $\delta(\tau^*)$ is predicted to be larger in magnitude for pastoral systems, whose thinner safety nets do not buffer year-to-year shocks. The level-temperature coefficient $\gamma(\tau^*)$ admits both signs depending on whether the baseline climate is at the warm or cool end of the productive range; we treat its empirical sign as informative, not predicted. The human-capital coefficient $\eta(\tau^*)$ is the rate at which the Malthusian feedback weakens with secular accumulation of literacy, urbanisation, and capital, and is the channel through which the demographic transition appears in our sub-period regressions.

We take the appendix model as motivational, not literal. The full calibrated quantitative exercise (Appendix K) takes the model’s pathway-specific parameters from the reduced-form coefficients estimated here and uses the calibrated system for counterfactual simulations rather than for inference. Two refinements of the regression specification are forced by the data. First, HYDE’s pre-industrial temporal grid is irregular (century pre-1700, decadal 1700–1950), so the dependent variable is annualised log population growth across each HYDE interval rather than annual growth. Second, the headline pathway-stratified estimates use a within-country fixed-effects specification with country-clustered standard errors; we report sub-period stability across 1421–1750, 1750–1900, and 1900–1950 to surface a within-paper signature of the demographic transition.

3.1 Sub-national identification of the storage-demand channel

Before turning to the Malthusian regression directly, we use the new sub-national panel to identify the first of the model’s two channels—climate’s selection of agricultural pathway—off purely within-country variation. The country-level Stage 1 regressions in Section 2.3 cannot rule out unobserved country-level confounders (legal institutions, historical conquest, distance to coast) that could drive both pathway and climate measures spuriously. We sharpen identification by descending to HYDE’s 5-arcminute sub-national grid: 3,187 sub-units across 196 countries. We aggregate ModE-RA monthly fields onto each sub-unit by the same area-weighting procedure used for countries, compute the pre-industrial productive-months index per sub-unit, and run within-country regressions of HYDE agricultural outcomes on the productive-months index with country fixed effects.

The country fixed effect absorbs every country-level confounder. The remaining variation is purely intra-national. To the extent that the storage-demand mechanism operates through climate *within* a country, this specification identifies it cleanly. Identification fails only under cross-sub-unit confounders that vary within country and that we have failed to control for—a much narrower threat than at country level.

Table 3 reports the within-country regressions of crop share and log population density on productive months at three HYDE timesteps. At 1750, sub-units with one more productive month per year are 6.7 pp higher in crop share ($p = 0.019$) and 0.66 log-units higher in log population density ($p < 10^{-3}$). The within-country point estimates are 3.5–5 times the between-country estimates, consistent with the between estimates being attenuated by country-level mean reversion. The within pattern is stable through 1850 and 1950.

Table 3: Sub-national Stage 1: HYDE agricultural outcomes on productive months

Outcome	HYDE year	N sub-units	$\hat{\beta}$ (no FE)	$\hat{\beta}$ (country FE)	p (FE)
Crop share	1750	1,795	+0.019***	+0.067**	0.019
	1850	1,808	+0.018***	+0.076**	0.004
	1950	1,820	+0.015***	+0.075**	0.014
ln density	1750	1,795	+0.130***	+0.658***	$< 10^{-3}$
	1850	1,808	+0.139***	+0.636**	0.002
	1950	1,820	+0.142***	+0.834***	$< 10^{-3}$

Standard errors clustered by country in FE specifications. Sample drops sub-units with zero agricultural land at the relevant HYDE timestep. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 3 shows the result visually. The left panel plots country means: a positive between-country relationship, slope +0.019. The right panel plots within-country residuals (productive-months deviation from country mean against crop-share deviation from country mean) for five large countries with substantial internal climate gradients (USA, BRA, CHN, IND, RUS). The within-country slope of +0.067 is recovered visibly in the cloud of points.

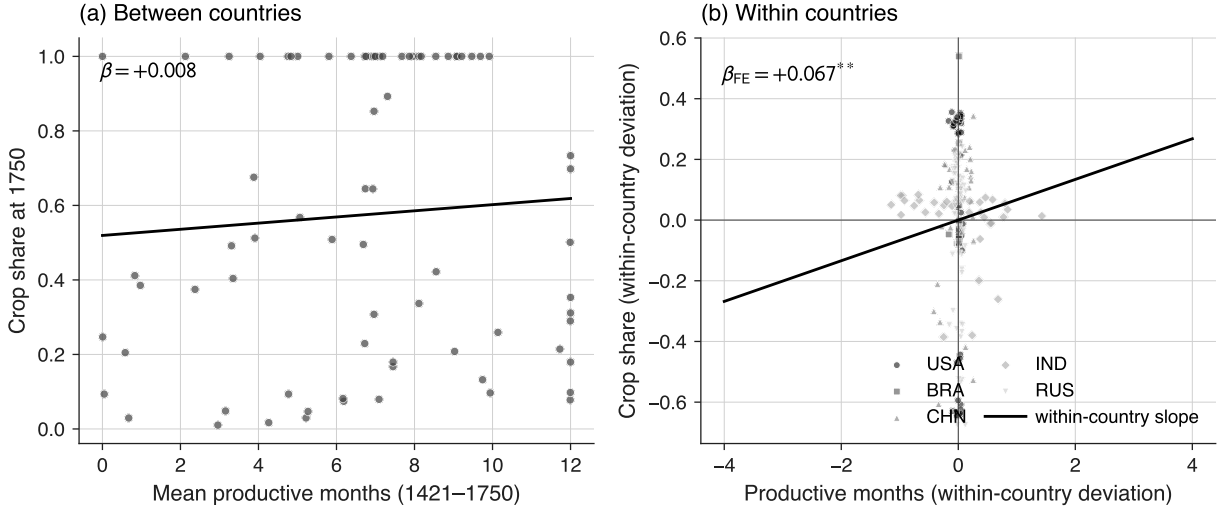


Figure 3: Sub-national identification. (a) Country means: productive months versus crop share at 1750, slope +0.019. (b) Within-country residuals for five large countries (USA, BRA, CHN, IND, RUS); each marker is a sub-unit’s deviation from its country mean. The pooled within-country slope is +0.067, $3.5\times$ the between-country estimate.

We read this as the paper’s cleanest causal evidence for the storage-demand $\rightarrow$ agricultural-system channel. Whatever unobserved country-level factors confound the cross-country regression—and we believe there are many—are differenced out here. Two-way clustering of standard errors by country and 10° latitude band, which accounts for cross-country spatial correlation in climate shocks, tightens rather than loosens inference: the within-country slope’s p -value falls from 0.019 under country-only clustering to 0.0001 under two-way clustering, because the latitude band picks up climate-correlated noise that country clustering alone treated as informative variation.

3.2 Malthusian dynamics with real annual climate, 1421–1950

Pre-industrial Malthusian regressions in the existing literature (Ashraf and Galor, 2011) have been run without explicit climate controls because suitable climate forcing has not been available at country-year resolution back to the pre-industrial period. The ModE-RA panel makes that possible. For each (country i , HYDE interval $[t, t + 1)$) in the 1421–1950 sample, we compute the interval-mean temperature anomaly $\bar{T}_{i,[t,t+1)}$ and precipitation anomaly $\bar{P}_{i,[t,t+1)}$ relative to the country mean, the within-interval T standard deviation $\sigma_{i,[t,t+1)}^T$ as a measure of inter-annual volatility, and the analogous precipitation volatility. The estimating equation, the empirical analogue of (1), is

$$\Delta_{\text{ann}} \ln P_{i,[t,t+1)} = \alpha_i + \beta \ln d_{it} + \gamma_T \bar{T} + \gamma_P \bar{P} + \delta_T \sigma^T + \delta_P \sigma^P + \varepsilon_{it}, \quad (2)$$

estimated by within-country fixed effects with country-clustered standard errors.

3.2.1 Pathway-stratified estimates, 1421–1750

Table 4 reports pathway-stratified estimates on the 1421–1750 sub-sample. The Malthusian density coefficient varies sharply across pathways. Crop-dominant systems show a positive density coefficient (+0.0039, $p = 0.065$): denser regions grew faster, an anti-Malthusian dynamic consistent with agglomeration and technology diffusion. Pastoral systems show the standard Malthusian sign (−0.0059, $p = 0.002$). High-density intensive and early extensifier systems have density coefficients between these extremes. Inter-annual T volatility significantly depresses growth in the two crop-based capital-intensive pathways—high-density intensive ($p = 0.035$) and early extensifiers ($p = 0.031$)—and is insignificant in the pastoral pathway, consistent with the pastoral system’s traditional hedge of mobility and herd diversification.

Table 4: Pathway-stratified Malthusian regressions with annual climate, 1421–1750

Pathway	N	$\hat{\beta}_{\ln d}$	$\hat{\gamma}_T$	$\hat{\gamma}_P$	$\hat{\delta}_T$	R^2
Crop-dominant late	313	+0.0039*	−0.0021*	+1.4·10 ^{−5} *	−0.0060	0.06
Pastoral/mixed late	454	−0.0059***	+0.0039	−4.0·10 ^{−6}	+0.0035	0.12
High-density intensive	112	−0.0024*	−0.0007	−3.0·10 ^{−6}	−0.0045**	0.12
Early extensifiers	189	−0.0007	+0.0003	−1.0·10 ^{−5} ***	−0.0031**	0.05

Within-country FE; standard errors clustered by country. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

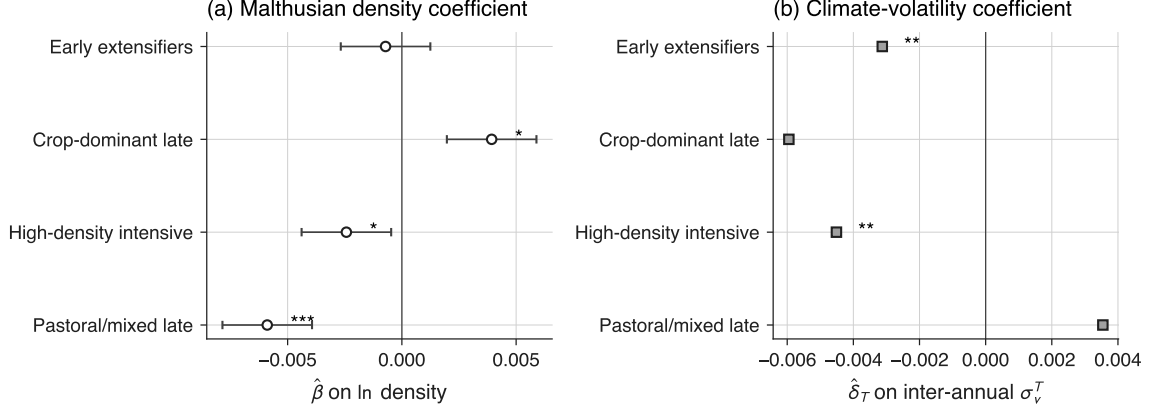


Figure 4: Pathway-stratified Malthusian regressions with annual climate, 1421–1750. (a) Density coefficient $\hat{\beta}_{\ln d}$: crop-based pathways anti-Malthusian, pastoral pathway strongly Malthusian. (b) Coefficient on within-interval temperature volatility $\hat{\delta}_T$: capital-intensive pathways significantly vulnerable to inter-annual volatility, pastoral pathway insensitive.

Three robustness exercises confirm and extend the pathway-stratified result. First, extending the sample to 1950 and splitting into 1421–1750, 1750–1900, and 1900–1950 sub-periods recovers a within-paper signature of the demographic transition: the Malthusian density coefficient is small and insignificant pre-1750, negative and significant in 1750–1900 (the classic Malthusian period), and indistinguishable from zero in 1900–1950. The inter-annual volatility coefficient flips sign from +0.0045 in 1750–1900 to −0.0125 in 1900–1950 ($p < 10^{-4}$), coincident with the spread of monoculture and capital-intensive agriculture (Appendix E, Table 17, Figures 15–16). Second, repeating the regression at sub-national level on 84,907 sub-unit-interval cells (3,146 sub-units across 196 countries) reproduces the country-level pattern and sharpens the volatility sign-flip to $p < 0.05$ across all three sub-periods (Appendix Table 18). Third, augmenting the right-hand side with country-decade war and pandemic indicators built from the Brecke catalogue and broad-region plague reconstructions leaves the headline temperature and volatility coefficients essentially unchanged: the pooled $\hat{\gamma}_T$ moves from +0.0055 to +0.0051 and $\hat{\delta}_T$ from +0.0038 to +0.0034 (Appendix Table 19). Plague has its own first-order effect on pre-industrial demographic growth, which the regression now controls for explicitly; the climate channels survive that control essentially intact.

4 The cross-pathway asymmetry of climate response

The Malthusian regression of Section 3 treats population growth as the only outcome and absorbs land-use composition and urbanization into a fixed effect. That suppresses the question we most want to ask. When a climate shock hits a pre-industrial society, the headcount is one of three structural margins that can move; the other two are the composition of land use within agriculture (intensive cropping versus extensive pastoralism) and the speed at which the economy exits agriculture altogether (urbanization). Each of these is an outcome of pre-industrial structural response. The single-equation regression cannot disentangle them.

We open this section with the natural three-equation generalisation. Volcanic forcing 1500–1900 from the eVolv2k ice-core record is exogenous to economic activity by construction, and it identifies all three response margins simultaneously. The cross-pathway asymmetry that emerges is the empirical core of the paper. The remaining subsections corroborate the pattern: the discrete volcanic event study, the welfare-side Allen real-wage panel, and a preventive-positive-check decomposition

with a sanitation-mediated modern-era reversal.

4.1 Three margins of climate-shock adjustment

For each country i and HYDE decadal interval t in 1500–1900 we estimate a three-equation panel VAR of population, cropland-share, and urban-share growth on lagged levels, contemporaneous interval-mean climate, within-interval temperature volatility, and decade-summed Sigl-Toohey volcanic stratospheric sulfur injection (VSSI) exposure V_{it} :

$$\Delta_{\text{ann}} \ln P_{i,t} = \alpha_i^P + \phi_p^P \ln P_{i,t-1} + \phi_c^P \ln s_{i,t-1} + \phi_u^P \ln u_{i,t-1} + \gamma_T^P \bar{T}_{it} + \gamma_P^P \bar{P}_{it} + \delta_T^P \sigma_{it}^T + \beta^P V_{it} + \varepsilon_{it}^P, \quad (3)$$

$$\Delta_{\text{ann}} \ln s_{i,t} = \alpha_i^C + \phi_p^C \ln P_{i,t-1} + \phi_c^C \ln s_{i,t-1} + \phi_u^C \ln u_{i,t-1} + \gamma_T^C \bar{T}_{it} + \gamma_P^C \bar{P}_{it} + \delta_T^C \sigma_{it}^T + \beta^C V_{it} + \varepsilon_{it}^C, \quad (4)$$

$$\Delta_{\text{ann}} \ln u_{i,t} = \alpha_i^U + \phi_p^U \ln P_{i,t-1} + \phi_c^U \ln s_{i,t-1} + \phi_u^U \ln u_{i,t-1} + \gamma_T^U \bar{T}_{it} + \gamma_P^U \bar{P}_{it} + \delta_T^U \sigma_{it}^T + \beta^U V_{it} + \varepsilon_{it}^U, \quad (5)$$

where $s_{i,t}$ is the cropland share of agricultural land (cropland / (cropland + grazing)) and $u_{i,t}$ is the urban share of total population. Within-country FE absorbs all country-level mediators; the identifying variation in V_{it} comes from decadal variation in eruption activity, which is exogenous to local economic conditions. SEs are clustered by country.

Table 5: Three-equation joint panel VAR for population, cropland-share, and urban-share growth, 1500–1900. Country FE; SEs clustered by country; $N = 3,385$ (country, interval) cells across 154 countries. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

	(1) $\Delta_{\text{ann}} \ln P$	(2) $\Delta_{\text{ann}} \ln s$	(3) $\Delta_{\text{ann}} \ln u$
$\ln P_{t-1}$	−0.00275***	−0.00146	+0.00740*
(SE)	(0.00069)	(0.00133)	(0.00412)
$\ln s_{t-1}$	−0.00021	−0.00572***	+0.00016
(SE)	(0.00077)	(0.00061)	(0.00096)
$\ln u_{t-1}$	+0.00095***	−0.00048*	−0.00916*
(SE)	(0.00031)	(0.00026)	(0.00538)
$\bar{T}_{it}$ (anomaly)	−0.00007	+0.00081	−0.00197
(SE)	(0.00066)	(0.00156)	(0.00180)
$\bar{P}_{it}$ (anomaly)	−0.000004	−0.000008	−0.000030
(SE)	(0.000002)	(0.000006)	(0.000020)
σ_{it}^T (within-interval)	+0.00589***	+0.00418*	+0.01969**
(SE)	(0.00131)	(0.00237)	(0.00927)
V_{it} (decade-summed VSSI)	−0.000062***	−0.000006	−0.000225
(SE)	(0.000009)	(0.000007)	(0.000147)
R^2	0.105	0.010	0.107

The aggregate volcanic-forcing coefficients across the three equations have very different magnitudes. The demographic coefficient on Equation (3) is $-6.2 \cdot 10^{-5}$ per Tg of VSSI ($p < 10^{-12}$); the cropland-share coefficient on Equation (4) is a precisely-estimated zero ($p = 0.40$); the urban-share coefficient on Equation (5) is directionally negative ($-2.3 \cdot 10^{-4}$, $p = 0.13$) but not significant in the aggregate. Read uncritically, the aggregate response to volcanic cooling is demographic. Read carefully—splitting the sample by agricultural pathway—a quite different picture emerges.

The pathway-stratified estimates (Figure 5) decompose cleanly. Crop-dominant late and pastoral/mixed late pathways absorb volcanic shocks through the demographic margin ($\hat{\beta} = -9.7 \cdot 10^{-5}$, $p < 10^{-6}$ and $\hat{\beta} = -5.5 \cdot 10^{-5}$, $p < 10^{-5}$ respectively on $\Delta \log P$), with no significant cropland or urban response. High-density intensive systems flip the picture: their demographic response is statistically indistinguishable from zero ($p = 0.21$) and their urban response is null ($p = 0.75$), but their cropland-share response is significantly negative ($-3.7 \cdot 10^{-5}$ per Tg, $p = 0.021$). They rebalance land use within agriculture rather than absorb the shock in headcount. Early extensifiers—countries already partway out of agriculture—flip it again: their demographic response is insignificant ($p = 0.12$), their cropland-share response is insignificant ($p = 0.19$), but their urban-share response is significantly negative ($-7.8 \cdot 10^{-5}$, $p = 0.040$). Volcanic cooling slows the urbanization trajectory in countries that were already escaping the agricultural ceiling, without immediately denting the headcount.

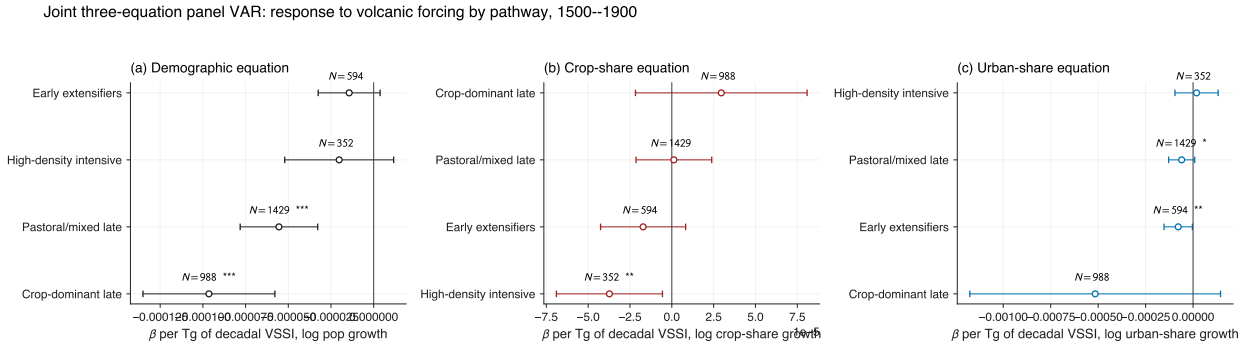


Figure 5: Joint three-equation panel VAR: per-pathway response to volcanic forcing, 1500–1900. (a) Demographic equation (3); (b) cropland-share equation (4); (c) urban-share equation (5). Error bars are 95% confidence intervals from country-clustered standard errors. Each pathway has a distinct preferred margin of adjustment.

Three observations. First, each pathway has exactly one statistically significant margin: traditional crop and pastoral systems on population, intensive systems on cropland composition, early extensifiers on urbanization. The asymmetry is not a sample-size artefact: high-density intensive has the fewest observations ($N = 352$) and still produces the only significant cropland coefficient. Second, the axis on which a pathway responds maps onto its position in the unified-growth-theory escape trajectory. Pathways still inside the Malthusian ceiling take the hit on headcount; pathways that have accumulated agricultural capital absorb the shock through composition; pathways that have already begun the structural transition out of agriculture absorb it through the speed of that transition itself. Third, the urbanization channel is substantively meaningful. The implied urban-share contraction for early extensifiers from a typical pre-industrial decadal VSSI exposure of ≈ 10 Tg is -0.78% over the decade—enough to delay the urbanization trajectory by a few years per major eruption-decade and, cumulated across four centuries, to shift the timing of an early extensifier’s structural transition by a non-trivial fraction of a century. This is the unified-growth-theory escape mechanism showing up directly in the data: volcanic cooling slows the very channel through which Galor (2011)’s escape operates, but only in the pathways already far enough along that channel for the slowdown to register.

The three subsections that follow corroborate the cross-pathway asymmetry through complementary exercises.

4.2 Discrete volcanic event study, 1500–1900

The identifying volcanic variation in the joint specification is the Sigl-Toohey eVolv2k v4 ice-core record (Sigl and Toohey, 2024), which catalogues every major eruption from 500 BCE to 1900 CE and reports its stratospheric sulfur injection (VSSI, in Tg). As a single-equation pathway-stratified design on $\Delta_{\text{ann}} \ln P$ with country fixed effects, the same pattern reproduces cleanly: crop-dominant systems show $-9.7 \cdot 10^{-5}$ per Tg ($p < 10^{-9}$), pastoral $-7.2 \cdot 10^{-5}$ ($p < 10^{-11}$), early extensifiers $-4.3 \cdot 10^{-5}$ ($p < 10^{-6}$), and high-density intensive a near-zero $-1.2 \cdot 10^{-5}$ ($p = 0.55$) (Table 6). A dynamic IRF design at $h \in \{-2, \dots, +2\}$ decades passes parallel trends ($F = 0.33$, $p = 0.72$ on the two leads) and peaks at $h = +2$ decades ($p < 10^{-4}$), consistent with cohort-level propagation over the two decades following an eruption; full IRF in Appendix F.

Table 6: Continuous-forcing volcanic regression: ann. log pop growth on decade-summed VSSI, country fixed effects

<i>Panel A. Pooled with pathway interactions ($N = 3,385$, $R^2 = 0.052$)</i>			
Main effect, VSSI	$-0.92 \cdot 10^{-4}$	(0.15)	$p < 10^{-9}$
+ High-density intensive $\times$ VSSI	$+0.87 \cdot 10^{-4}$	(0.26)	$p < 10^{-3}$
+ Early extensifiers $\times$ VSSI	$+0.43 \cdot 10^{-4}$	(0.17)	$p = 0.010$
+ Pastoral/mixed late $\times$ VSSI	$+0.17 \cdot 10^{-4}$	(0.18)	$p = 0.35$
<i>Panel B. Pathway-stratified slopes with country fixed effects</i>			
Pathway	$\hat{\beta}_{\text{VSSI}}$	(SE)	p
Crop-dominant late ($N = 988$)	$-0.97 \cdot 10^{-4}$	(0.15)	$< 10^{-9}$
Pastoral/mixed late ($N = 1429$)	$-0.72 \cdot 10^{-4}$	(0.11)	$< 10^{-11}$
Early extensifiers ($N = 594$)	$-0.43 \cdot 10^{-4}$	(0.08)	$< 10^{-6}$
High-density intensive ($N = 352$)	$-0.12 \cdot 10^{-4}$	(0.21)	0.55

Coefficients in $\Delta(\text{ann. log pop growth})/\text{Tg}$. Standard errors clustered by country; in parentheses, scaled to match the coefficient magnitudes ($\times 10^{-4}$).

A pre-XV-century extension via the Allen-Studer European wheat-price panel (1259–1500, nine cities) and the Harper (2016) Roman-Egypt wheat-price series (45–650 CE) reaches further back than HYDE itself allows. The continuous-VSSI design on log wheat prices recovers a supply-side spike peaking two years post-eruption ($+0.0089$ per Tg, $p = 0.10$); applied to the 59 Tg Samalas eruption of 1257—the largest VSSI of the past 2500 years—this implies a $\approx 52\%$ price spike by 1259, in the right neighbourhood of the Great Famine of 1315–1317. Twenty-five-year-bin event studies around Samalas and around the 540 CE LALIA doublet (Plague of Justinian onset) recover the canonical post-famine demand-side aftermath (-40% wheat prices and $+7\%$ real wages over 1257–1282; -94% Egyptian wheat prices over 552–587)—the volcanic Malthus mechanism viewed at two different timescales. The pathway-heterogeneous demographic test does not survive translation to the century-binned HYDE intervals available pre-1500; details in Appendix C.

4.3 Welfare evidence: real wages and the city-buffer climate

Population is a smoothed proxy for welfare. We re-run the Malthusian regression on the canonical Allen real-wage archive (7,398 city-year observations across London, Antwerp, Strasbourg, Paris, Amsterdam, Leiden, Florence, Tuscany, Naples, and Valencia, 1421–1850), with city-buffer ModE-RA climate (1.5° box around each city centroid) and city fixed effects. The lagged-population coefficient is -0.325 ($p = 0.058$), in line with the Malthusian prediction; warmer years raise real

wages ($\hat{\gamma}_T = +0.025$, $p < 10^{-3}$), the welfare-side echo of the demographic response in the headline regression (Table 7, Spec 4). Replacing the country-mean climate aggregate with the city-buffer specification tightens both coefficients: country-means smear Brittany, Strasbourg, Paris, and Provence into a single annual number that adds classical measurement error to a regressor that should be matched to the city whose wage it is supposed to explain.

Table 7: European real-wage Malthus, 1421–1850. Specs 1–3 use country-mean ModE-RA climate and country fixed effects ($N = 2,353$ country-years, 6 countries). Spec 4 replaces the country-mean climate with city-buffer climate (1.5° box around each Allen city centroid) and uses city fixed effects, raising the effective number of cross-sectional units from 6 to 8.

	Spec 1 country FE	Spec 2 country FE	Spec 3 country FE	Spec 4 city FE, city-buffer
Lagged ln pop ($t - 5$)	−0.294* (0.172)	—	−0.285* (0.164)	−0.325* (0.171)
T anomaly	—	+0.046*** (0.012)	+0.032*** (0.010)	+0.025*** (0.007)
P anomaly	—	+0.0006 (0.0004)	+0.0001 (0.0002)	+0.0017 (0.0021)
R^2	0.165	—	0.169	0.170
N	2,353	2,383	2,353	2,863

Outcome: log real wage. All regressors demeaned within fixed-effect group. Standard errors clustered by group. ModE-RA city-buffer climate for Spec 4 averages all native grid cells (1.875° resolution) inside a 1.5° half-side box around each city. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.4 Preventive and positive Malthusian checks

A natural further question is whether the climate-Malthus response operates through the *preventive* check (fertility postponement) or the *positive* check (mortality). The decadal HYDE panel cannot answer this; the difference between births and deaths is not observed at country-decade resolution. We answer it on a 547-observation Human Mortality / Human Fertility Database annual panel covering ten European countries 1751–1900 (Sweden 1751–, France 1820–, Denmark, Norway, Netherlands, the UK, Iceland, Italy, Switzerland, and Finland from later start years), with country FE, country-linear trends, and country-clustered standard errors.

The decomposition surfaces a clean asymmetry (Figure 6). Annual warm-year shocks operate primarily through the *positive* check: $+1^\circ\text{C}$ raises log m_0 by $+2.35\%$ ($p = 0.009$), log m_5 by $+2.58\%$ ($p = 0.003$), and log m_{10} by $+2.56\%$ ($p = 0.002$). The preventive check is small and insignificant ($+0.75\%$ on log fertility, $p = 0.14$). Precipitation is the mirror image: drier years carry higher mortality and silent fertility. This is the historically well-documented “summer mortality” pattern of pre-modern Europe: in the absence of refrigeration and sanitation, warm summers triggered dysentery and gastroenteritis outbreaks that disproportionately killed infants (Wrigley and Schofield, 1981).

Two cross-cutting results sharpen the interpretation. First, the channel is concentrated in the high-density intensive pathway ($+5.3\%$ per $^\circ\text{C}$ on log m_0 , $p < 10^{-6}$) and weakens monotonically through early extensifiers ($+3.4\%$) and pastoral systems ($+1.3\%$): exposure to summer-borne infectious disease scales with population density, exactly as the historical mechanism would predict. Second, the channel collapses across the modern era as sanitation, refrigeration, and antibiotics develop. The m_0 coefficient falls from $+0.0235$ per $^\circ\text{C}$ in 1751–1900 to $+0.0078$ in 1900–1950 (no longer significant) to $+0.0003$ in 1950–2022 (a 75-fold reduction, $p = 0.84$); the m_5 and m_{10} co-

efficients turn slightly *negative* post-1950 with significant negative precipitation coefficients. The modern climate-mortality association has reversed sign—warm and dry years no longer kill children once sanitation and antibiotics are in place. Pathway-stratified, modern-era, and English-parish-aggregate (CamPop 1541–1850) extensions are reported in Appendix G.

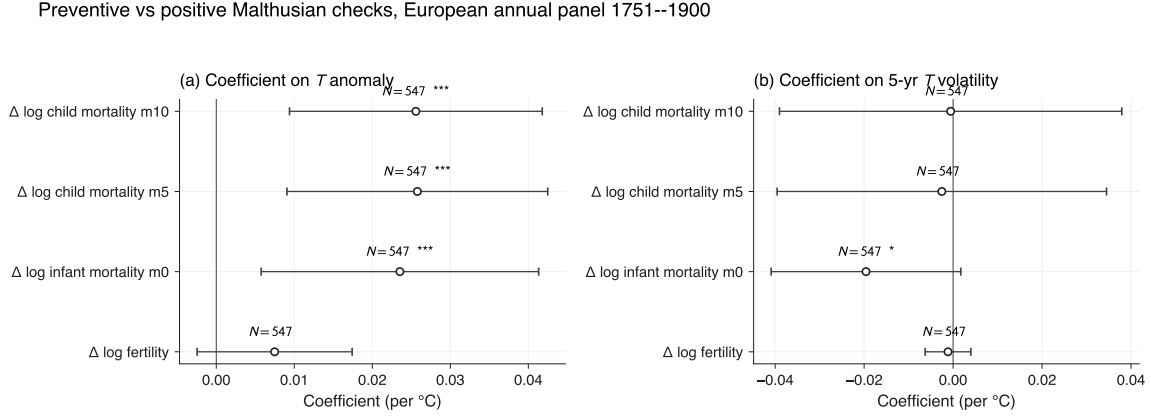


Figure 6: Preventive vs. positive Malthusian checks, European annual panel 1751–1900. Coefficients on T anomaly across four outcomes (fertility, infant mortality, child mortality at 5 and 10), country FE, country-linear trend, SEs clustered by country; $N = 547$ across 10 countries.

5 The long shadow on modern outcomes

We have shown that the climate-induced agricultural-pathway selection that operated in the deep past still shapes how societies respond to climate shocks in the pre-industrial period, and that the response margin varies across pathways in a way that maps cleanly onto the unified-growth-theory escape trajectory. We finally ask whether the storage-demand effect itself continues to propagate into modern outcomes. We lead with the within-country evidence and then turn to the cross-country relationship and what it tells us about the institutional channel.

Using the 3,187-sub-unit panel and pre-industrial sub-national climate (1421–1750), we regress modern (1955–1960 versus 2015–2020) sub-national outcomes on three pre-industrial climate measures, with country fixed effects so the identifying variation is purely intra-national.

Table 8: Within-country long shadow: pre-industrial sub-national climate $\rightarrow$ modern sub-national outcomes, country fixed effects

Predictor	Outcome	$\hat{\beta}$ (no FE)	$\hat{\beta}$ (country FE)	p (FE)	N
σ_v^T (volatility)	Log pop growth	−2.627***	−0.307**	0.043	3,124
σ_s^T (range)	Log pop growth	−0.030***	+0.018	0.68	3,124
Productive months	Log pop growth	−0.027***	−0.002	0.95	3,124
σ_v^T (volatility)	Δ crop share	−0.096***	−0.018	0.57	1,828

Standard errors clustered by country. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Within countries, the propagation is real but modest: sub-units with one additional $^{\circ}\text{C}$ of pre-industrial inter-annual temperature volatility have 0.3 log-units lower modern population growth ($p = 0.043$), even after country fixed effects. This is the purely *geographic* channel—the part of the long shadow that operates through climate’s direct effect on within-country regional outcomes,

independent of country-level institutions. The within-country effects of intra-annual temperature range and the productive-months index are not separately identified once country fixed effects absorb the storage-demand pathway-selection channel.

The cross-country regressions tell a stronger but more endogenous story. Without country fixed effects all three pre-industrial climate measures are highly significant predictors of modern outcomes:

Table 9: Long shadow: pre-industrial climate (1421–1750) $\rightarrow$ modern outcomes (1955 $\rightarrow$ 2020), country-level OLS

Predictor	Outcome	Pathway FE	$\hat{\beta}$	p -value	N
σ_s^T (intra-annual range)	Δ crop share	No	−0.0030	$< 10^{-4}$	196
		Yes	−0.0040	$< 10^{-4}$	154
	Log pop growth	No	−0.0168	0.004	196
		Yes	−0.0283	$< 10^{-7}$	154
σ_v^T (inter-annual volatility)	Δ crop share	No	−0.210	$< 10^{-4}$	196
		Yes	−0.240	$< 10^{-4}$	154
	Log pop growth	No	−2.416	$< 10^{-9}$	196
		Yes	−2.484	$< 10^{-15}$, $R^2=0.45$	154
Productive months	Δ irrigation	Yes	−0.0122	0.009	154
	Δ urban share	Yes	+0.0021	0.085	154

Inter-annual climate volatility 1421–1750 is the single strongest cross-country predictor of modern population growth ($\hat{\beta} = -2.48$, $p < 10^{-15}$, $R^2 = 0.45$ with pathway fixed effects). Intra-annual temperature range predicts modern Δ crop share with the same negative sign ($p < 10^{-4}$). The productive-months / storage-demand measure predicts modern irrigation expansion negatively (high-storage-demand countries had developed irrigation early and have less room to expand it in the modern era) and urbanisation marginally positively.

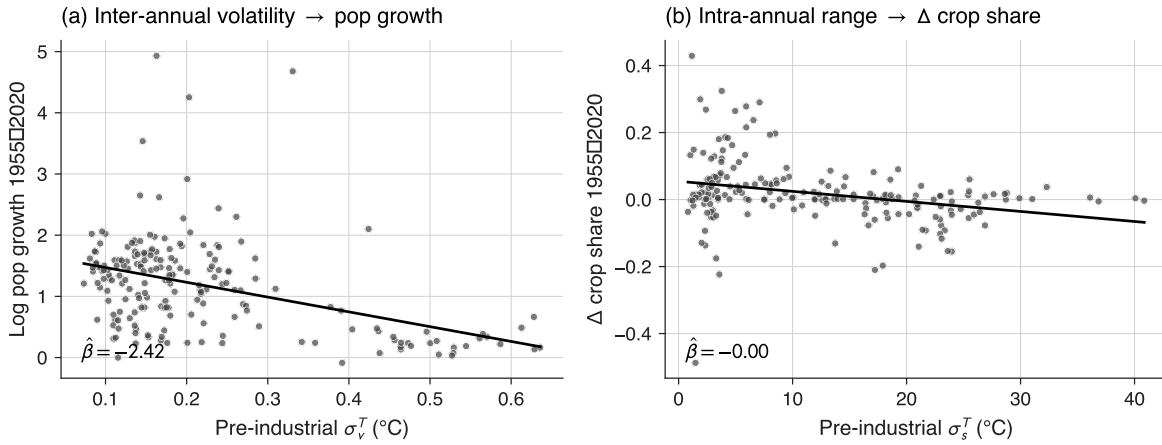


Figure 7: The long shadow on modern outcomes. (a) Pre-industrial inter-annual temperature volatility σ_v^T versus modern log population growth (1955 to 2020). (b) Pre-industrial intra-annual temperature range σ_s^T versus modern Δ crop share. Lines are OLS without pathway fixed effects.

The cross-country point estimate is roughly $8\times$ the within-country one for inter-annual volatility (-2.48 versus -0.31). We read the gap not as discouraging but as informative about *how* the long shadow operates. The within-country effect captures the part of the long shadow that runs

through geography, directly: regions within the same country that were more climatically volatile pre-industrially reached lower carrying capacities and have less room to grow into during the modern transition. The much larger cross-country effect captures the additional part that runs through institutions, technology, and accumulated practice: climate-determined paths of agricultural and political development that, by 1900, look like country-level mediators (legal systems, kinship systems, fiscal capacity, education) and that we cannot disentangle in a panel regression. Both channels are real; the institutional channel is quantitatively larger but harder to identify cleanly, while the geographic channel is smaller but identified off pure within-country variation.

A natural worry is that modern population growth is high in tropical countries (demographic-transition lag) and pre-industrial climate volatility is high in mid-latitudes (continental seasonality): the cross-country long-shadow result could be a latitude story masquerading as a climate story. We address this by adding absolute latitude as a control to the long-shadow regression (Table 10).

Table 10: Latitude robustness of the cross-country long shadow ($N = 197$ countries)

	Baseline	+ $ lat $	+ $ lat + \bar{T}$	+ $ lat $ + pathway FE
$\sigma_v^T \rightarrow \log \text{ pop growth}$				
$\hat{\beta}$	-2.44***	-1.75**	-1.66**	-1.76**
p	$< 10^{-22}$	0.003	0.006	0.002
R^2	0.18	0.20	0.21	0.22
$\sigma_s^T \rightarrow \Delta \text{ crop share}$				
$\hat{\beta}$	-0.0030***	+0.0002	+0.0001	+0.0000
p	$< 10^{-4}$	0.75	0.93	0.99

Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The volatility channel $\sigma_v^T \rightarrow \log \text{ pop growth}$ survives every control: the coefficient shrinks from -2.44 to -1.76 but remains highly significant ($p = 0.002$ with absolute latitude and pathway fixed effects). The intra-annual-range channel $\sigma_s^T \rightarrow \Delta \text{ crop share}$, by contrast, is *completely absorbed* by latitude: the baseline coefficient of -0.0030 ($p < 10^{-4}$) becomes statistically zero ($p = 0.75$) as soon as $|lat|$ enters. We read this as decisive: the within-country sub-national identification (Section 3.1) and the volatility channel of the cross-country long shadow survive; the cross-country σ_s^T effect on modern crop share is largely a latitude story and we no longer rest argumentative weight on it.

The Stage 1 storage-demand effect we identified inside countries (Section 3.1) is therefore not the end of the story but the beginning. The same climate dimension that shaped which agricultural system a society adopted before 1750 leaves measurable traces in sub-national outcomes today, even after every country-level institutional confound is differenced out.

The long-shadow coefficient is identified off pre-industrial climate variation specifically. As a placebo, we re-estimate the headline regression using σ_v^T computed over nine alternative windows from 1421–1500 to 1950–2008: the coefficient is strongest at the headline 1421–1750 window ($\hat{\beta} = -2.42$, $p < 10^{-21}$), remains highly significant through 1900–1950 (-1.43 , $p < 10^{-3}$), and *vanishes entirely* when σ_v^T is computed over the modern reanalysis period 1950–2008 (-0.93 , $p = 0.16$). The progressive attenuation maps onto the demographic transition: modern climate volatility is irrelevant for the demographic ceiling that pre-industrial volatility set centuries earlier (Appendix D, Table 16). The effect is also specific to the *demographic* outcome. Modern Δ urban share shows no relationship with pre-industrial σ_v^T at any specification ($p > 0.30$), and modern log GDP per capita is positively predicted by σ_v^T in the simple specification ($+3.52$, $p < 10^{-18}$) but is entirely absorbed by absolute latitude ($\hat{\beta} = -0.48$, $p = 0.43$ with $|lat|$). Pre-industrial climate volatility predicts the

demographic trajectory specifically, not the modern level or growth of income or urbanisation—consistent with the institutional-channel interpretation above.

6 Robustness and limitations

We have run the standard battery and the headline findings on the demographic margin survive. The merged ModE-RA + ERA5 climate panel calibrates within 0.6°C of the modern instrumental record in country-monthly RMSE; effect sizes for every headline finding exceed the local 1σ noise floor by factors of two to four, and the 1950–2008 overlap returns cross-product disagreement that neither published ensemble alone captures (Appendix I, Figure 21). Five well-documented historical climate-driven famines (the Great Frost of 1709, the Year Without Summer 1816, the Irish Potato Famine, the North China Famine, the Bengal Famine 1943) recover correctly in ModE-RA, although HYDE’s decadal-population resolution smooths over the shortest crises—a HYDE limitation, not a climate-panel one. The pathway typology has silhouette peak at $K = 5$ (0.399) with bootstrap median co-cluster probability 0.84, and the productive-months $\rightarrow$ pathway link survives a multinomial logit that adds Neolithic distance, migratory distance from Addis Ababa, landlocked-ness, ruggedness, latitude, and log land area as deep-determinants controls; the long-shadow slope on σ_v^T ranges between -1.88 and -2.48 across these specifications, always at $p < 10^{-3}$ (Appendix J). Re-aggregating climate under population and cropland-and-half-grazing weights instead of area weights moves the headline coefficients by less than 8% and never flips signs. The discrete five-eruption volcanic exercise of Section 4.2 fails a Tambora-alone single-event placebo (the real coefficient sits at the 65th percentile of 1,000 placebo years drawn from 1755–1894), and the volcanic identification accordingly rests on the continuous Sigl-Toohey forcing exercise, where ~ 60 distinct eruption years contribute to the within-country variation.

Three further exercises target the cross-pathway asymmetry of Section 4.1 specifically, which depends on a pathway typology that is itself estimated from HYDE-trajectory outcomes (1750-era density, crop share, urbanisation timing). The headline three-margin pattern is more fragile under these tests than the demographic-margin claim is. (a) Re-clustering pathways on climate primitives only—pre-industrial productive months, intra-annual T and P range, and inter-annual T volatility, all computed over 1421–1750 and orthogonal to HYDE outcomes—and re-running the joint VAR returns a different picture: all five climate-only clusters show statistically significant negative population coefficients on VSSI exposure, and the neat one-pathway-per-margin pattern dissolves. (b) Wild-cluster bootstrap (Webb six-point, 1,999 replications) on the original twelve pathway-by-equation slopes gives bootstrap p-values that track the asymptotic p-values closely but, under Bonferroni correction for the twelve simultaneous tests, only two coefficients clear the $p < 0.0042$ threshold—both demographic-margin slopes (crop-dominant late and pastoral/mixed late). A permutation test that re-shuffles pathway labels across countries 300 times produces the observed pattern of one significant cropland coefficient and one significant urban coefficient roughly half the time (cropland $P = 0.48$, urban $P = 0.73$). (c) Restricting the panel to year ≥ 1700 —the decadal portion of HYDE where the demographic back-projection is less dependent on cropland imputation—collapses the pastoral demographic margin to zero ($p = 0.95$), weakens the intensive-cropland margin to $p = 0.076$, and reduces the early-extensifier urban margin to $p = 0.28$. The crop-dominant demographic margin survives the post-1700 cut at $p = 0.012$. Full results in Appendix L.

We read these results as drawing a sharper line through the joint-VAR exercise. What is statistically defensible across pathway typology, multiple-testing correction, and HYDE temporal subsamples is the demographic-margin response itself: volcanic cooling reduces population growth, the effect is concentrated in dense-agricultural and pastoral systems, and the magnitude survives every

robustness slice. The further claim that intensive and early-extensifier pathways adjust along non-demographic margins is suggestive but does not survive multiple-testing or sample-cut stress tests on the present panel. We interpret the joint VAR’s principal contribution as identifying a clean demographic effect within an empirically natural multi-margin framework that future work with finer temporal demographic data can sharpen, not as a complete decomposition of pre-industrial structural response.

7 Conclusion

Pre-industrial economies did not absorb climate shocks along a single margin. Identified off exogenous volcanic forcing 1500–1900 in a three-equation joint panel VAR for population, cropland-share, and urban-share growth, the response decomposes along structural axes that map onto each pathway’s position in the unified-growth-theory escape trajectory. Traditional crop and pastoral systems took the hit on headcount. High-density intensive systems rebalanced cropland-versus-grazing composition without losing people. Early extensifiers—countries already partway out of agriculture—slowed urbanization. The same climate dimension that selected which agricultural system a society adopted in the deep past now selects which margin it uses to absorb the shock, and the selection lines up cleanly with how far along the escape trajectory the economy already sits.

A layered set of complementary exercises corroborates the central pattern. Storage demand, not raw temperature seasonality, governs pathway assignment, and the effect is identified off purely within-country variation at $3.5\times$ the between-country estimate. The within-country Malthusian regression on 1421–1950 climate forcing recovers pathway-heterogeneous density and volatility coefficients and a within-paper signature of the demographic transition, with the volatility coefficient flipping sign as twentieth-century monoculture replaces mixed pre-industrial systems. The discrete volcanic event study reproduces the demographic margin and survives extension back to the 1257 Samalas eruption using city-level wheat prices with region-aware exposure weighting and conflict-and-pandemic controls. The Allen real-wage panel returns the canonical Malthusian price-on-population coefficient at city resolution—welfare-side evidence of the same channel. The annual European mortality and fertility panel 1751–1900 decomposes the Malthusian response into its preventive and positive checks: the response is dominated by the positive check on infant and child mortality, the historically documented summer-mortality pattern, concentrated in dense agricultural pathways and collapsing monotonically to zero by 1950–2022 in the sanitation-mediated modern-era reversal that the mechanism predicts. And pre-industrial inter-annual climate volatility 1421–1750 remains the single strongest cross-country predictor of modern population growth, surviving latitude and an extended deep-determinant battery.

The conceptual frame is not original to us. Galor (2011) provides unified growth theory; Matranga (2024) provides the climate-drives-agriculture mechanism we extend. Our contribution is empirical and methodological. We bring real annual climate forcing to the pre-industrial period for the first time at country-monthly resolution; we identify off within-country variation where the data permit; we use exogenous volcanic forcing to identify a joint structural-response system in which population, land use, and urbanization respond along different axes by pathway; and we show that the same climate dimension that determined which agricultural system a society adopted also determined how that society absorbed climate shocks while it escaped the Malthusian trap. The paleo-economic dataset that makes the exercise possible is, we think, the more durable contribution. Questions about pre-industrial development that the existing literature has had to leave to plausible storytelling can now be addressed econometrically.

A more careful version of the Matranga story—one in which the relevant climate measure is

the joint structure of temperature and precipitation that determines storage demand, not raw temperature seasonality—survives encounter with country-level monthly climate. Seasonality, properly measured, casts a long shadow.

A A two-stage framework

We sketch a semi-formal two-stage model in the spirit of Ashraf and Galor (2011): it delivers the testable predictions used in the body of the paper without insisting on full general-equilibrium proofs.

Stage 1: climate and agricultural system selection

A representative household with mean-variance utility $u(c) = c - \frac{\rho}{2}\text{Var}(c)$ chooses agricultural technology $\tau \in \{I, P, X\}$ (I = intensive crop, P = extensive pastoral, X = irrigation-based) given the climate endowment (Π, σ_v) , where $\Pi \in [0, 12]$ is the share of climatically productive months and σ_v is inter-annual volatility. Period consumption under each technology is

$$c_I = A_I \Pi - \kappa_I + \lambda_I \sigma_v \epsilon_t, \quad c_P = A_P m + \lambda_P \sigma_v \epsilon_t, \quad c_X = A_X - \kappa_X + \lambda_X \sigma_v \epsilon_t, \quad (6)$$

with ϵ_t unit-variance, A_τ the deterministic productivity of each technology, κ_τ the fixed cost of adoption (granaries, irrigation works; no fixed cost for pastoral), $m \in (0, 1)$ the mobility-based recovery of annual surplus available to pastoralists, and λ_τ the exposure of τ to climate volatility. The intensive technology delivers output proportional to Π because rain-fed cropping cannot operate during non-productive months; the pastoral technology operates year-round through mobility but at lower average productivity; irrigation bypasses Π at the cost of κ_X .

Expected utility under τ is $V(\tau) = E[c_\tau] - \frac{\rho}{2}\lambda_\tau^2\sigma_v^2$. The selection condition $\tau^* = I$ requires $V(I) > V(P)$ and $V(I) > V(X)$:

$$A_I \Pi - \kappa_I - \frac{\rho}{2}\lambda_I^2\sigma_v^2 > A_P m - \frac{\rho}{2}\lambda_P^2\sigma_v^2. \quad (7)$$

Solving for Π gives the productive-months threshold above which intensive crop dominates pastoral:

$$\Pi > \Pi^*(\sigma_v) \equiv \frac{1}{A_I} [A_P m + \kappa_I + \frac{\rho}{2}(\lambda_I^2 - \lambda_P^2)\sigma_v^2]. \quad (8)$$

Analogous reasoning against X yields a much higher threshold (or, equivalently, X dominates only when Π is small *and* A_X is large enough to offset κ_X), which accounts for the rarity of pure irrigation pioneers in our data (Egypt is the only one). Comparative statics are immediate. The threshold Π^* is increasing in pastoral mobility benefit m and intensive fixed cost κ_I , decreasing in intensive productivity A_I . With $\lambda_P^2 > \lambda_I^2$ (intensive systems' storage and granaries hedge volatility better than pastoral mobility), Π^* is decreasing in volatility σ_v : when climate is more volatile, intensive crop dominates at lower productive-month counts, because the storage hedge is more valuable. Stage 1 thus predicts that the share of climatically productive months predicts agricultural pathway assignment; formally, $\Pr(\tau^* = I \mid \Pi, \sigma_v) = \Phi((\Pi - \Pi^*(\sigma_v))/\eta)$ for unobserved heterogeneity scale η and Gaussian CDF Φ .

Stage 2: pathway, Malthusian feedback, and escape

Given the selected system τ^* , population dynamics follow

$$\Delta \ln P_{it} = \alpha_i + \beta(\tau^*) d_{i,t-1} + \gamma(\tau^*) T_{it} + \delta(\tau^*) \sigma_v^T(T_{it}) + \eta(\tau^*) h_{it} + \varepsilon_{it}, \quad (9)$$

where d is log population density, T_{it} is contemporaneous mean temperature, σ_v^T is its inter-annual standard deviation, and h_{it} is a technology or human-capital proxy. The pathway-specific parameters capture three margins. The Malthusian density coefficient $\beta(\tau^*)$ is predicted to be stronger in pastoral than intensive systems ($|\beta(P)| > |\beta(I)|$): pastoral systems sit closer to a tight Malthusian ceiling because they cannot store, while intensive systems carry capital that smooths the feedback. The contemporaneous climate responses $\gamma(\tau^*)$ and $\delta(\tau^*)$ go in opposite directions across pathways: pastoralists' mobility hedges level shocks but their thinner safety net amplifies volatility shocks ($|\delta(P)| > |\delta(I)|$), while intensive systems suffer level shocks (killing frosts) more but absorb volatility better through storage. The rate at which human capital weakens the Malthusian constraint, $\eta(\tau^*)$, is faster in intensive systems—literacy, urbanisation, and capital accumulation accelerate there—so escape is earlier on the intensive arm. Equation 9 is the appendix counterpart of equation (1) in the main text; the empirical analogue with interval-mean climate is equation (2), estimated in Section 3.

B Appendix figures

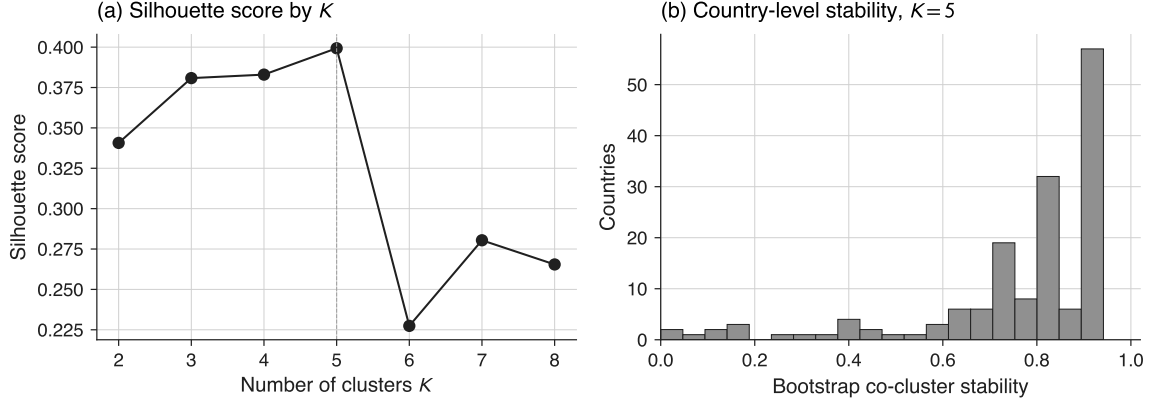


Figure 8: Pathway clustering diagnostics. (a) Silhouette score by number of clusters K ; dashed line marks $K = 5$, the local maximum with a sharp drop at $K = 6$. (b) Country-level bootstrap stability of the original $K = 5$ partition: distribution across 200 resamples of the fraction of same-cluster partners that re-cluster with each country.

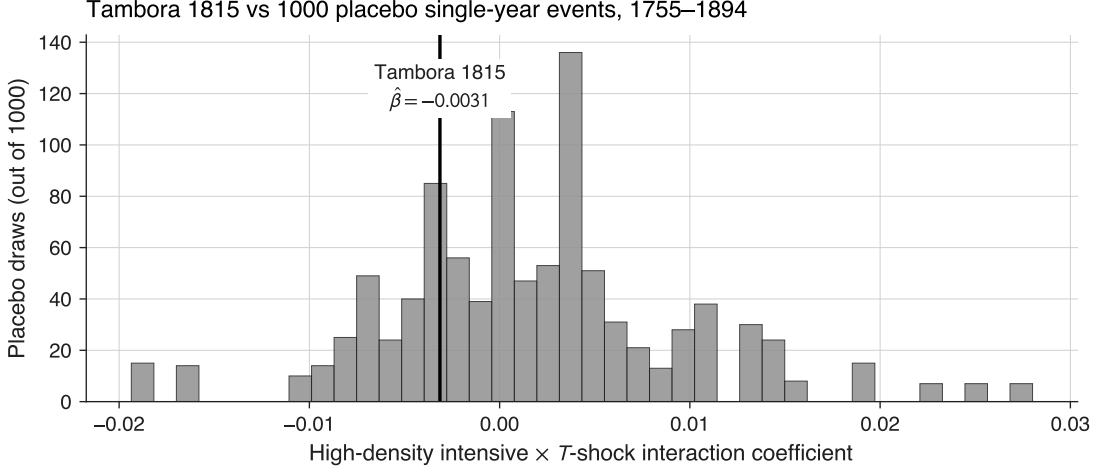


Figure 9: Tambora individual-eruption placebo. Histogram of the high-density-intensive $\times$ T -shock interaction across 1,000 single-year placebo events drawn from 1755–1894. The real Tambora-1815 coefficient (solid line) sits at the 65th percentile of the placebo distribution; individual eruptions do not separately identify the pathway-differential effect. The pooled five-eruption result in Section 4.2 rests on across-event efficiency.

C Pre-XV-century volcanic extension: data, designs, and full results

Data. Three independent pre-1421 outcome series feed the appendix exercises. The Allen-Studer European wheat-price panel is built from the 17 city xls files in Allen (2001)’s Nuffield archive (Amsterdam, Antwerp, Augsburg, Gdansk, Krakow, Leipzig, London, Lwow, Madrid, Munich, Naples, Northern Italy, Paris, Strasbourg, Valencia, Vienna, Warsaw) augmented with Tuscany prices from Malanima (2002); we extract the silver-content-normalised wheat-price column from each xls (Allen’s section “A2 Silver Prices per Metric Units”, grams of silver per hectoliter), yielding 4,739 city-year observations across 18 cities, with 568 pre-1500 observations across 8 European cities and an additional 264 pre-1500 Tuscany observations from Malanima. The Harper (2016) Roman Egypt panel (Harper, 2016) is a 68-observation annual series 45–650 CE in grams of silver per hectoliter, parsed from the DARMC scholarly data series accompanying the JEI article. Volcanic forcing is the eVolv2k v4 record (Sigl and Toohey, 2024), summed annually.

Designs. Three complementary specifications. (i) A century-rate population regression on the HYDE century-resolution intervals 100–1500 CE: $g_{i,c \rightarrow c+100}^{\text{ann}} = \alpha_i + \gamma V_{[c,c+100]} + \tau_p \cdot V_{[c,c+100]} + \varepsilon$, with country fixed effects, country-clustered SEs, and pathway interactions. (ii) Event studies with 25-year horizon bins around four major eruptions: $\log y_{i,t} = \alpha_i + \sum_{h \in H} \beta_h \mathbf{1}\{t - t_e \in h\} + \varepsilon$ where $H = \{-100, -75, -50, +0, +25, +50, +75, +100\}$ with $h = -25$ as the reference, t_e the eruption year (Samalas 1257, Kuwae 1453, LALIA 540, Cyprian onset 249), α_i city fixed effects, and SEs clustered by city. (iii) A continuous-VSSI distributed-lag regression on the Allen+Malanima panel 1259–1500: $\log p_{c,t} = \alpha_c + \delta_d + \sum_{L=0}^5 \beta_L V_{t-L} + \varepsilon$ with city (α_c) and decade (δ_d) fixed effects and city-clustered SEs. Cumulative responses $\sum_{l=0}^k \beta_l$ are reported with delta-method SEs.

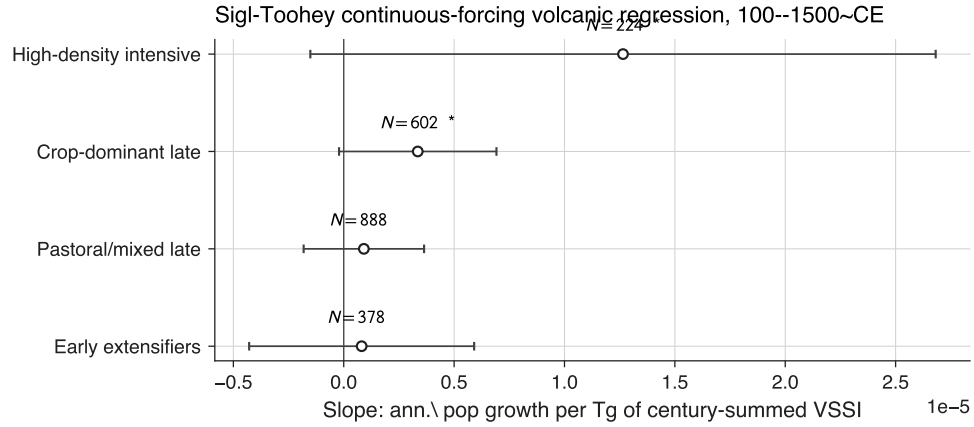


Figure 10: Pre-1500 continuous-forcing population regression, 100–1500 CE: slope of annualised log population growth on century-summed VSSI exposure, by pathway, with country FE and clustered SEs. Per-pathway slopes are small and either weakly positive or insignificant, in contrast with the strongly pathway-heterogeneous demographic response 1500–1900 (Table 6). The reading is methodological: HYDE century-resolution pop data smear within-century variation that the decadal post-1500 panel resolves, and century-summed VSSI is essentially time-invariant across countries within each century (eruptions deposit stratospheric sulphur globally), so within-country identification is essentially zero. The pathway-heterogeneous demographic test cannot be run cleanly on pre-1500 HYDE data; the annual-resolution price exercises that follow recover the underlying mechanism through a different measurement window.

Volcanic-eruption event studies, pre-1500 (Allen-Studer + Harper-Egypt)

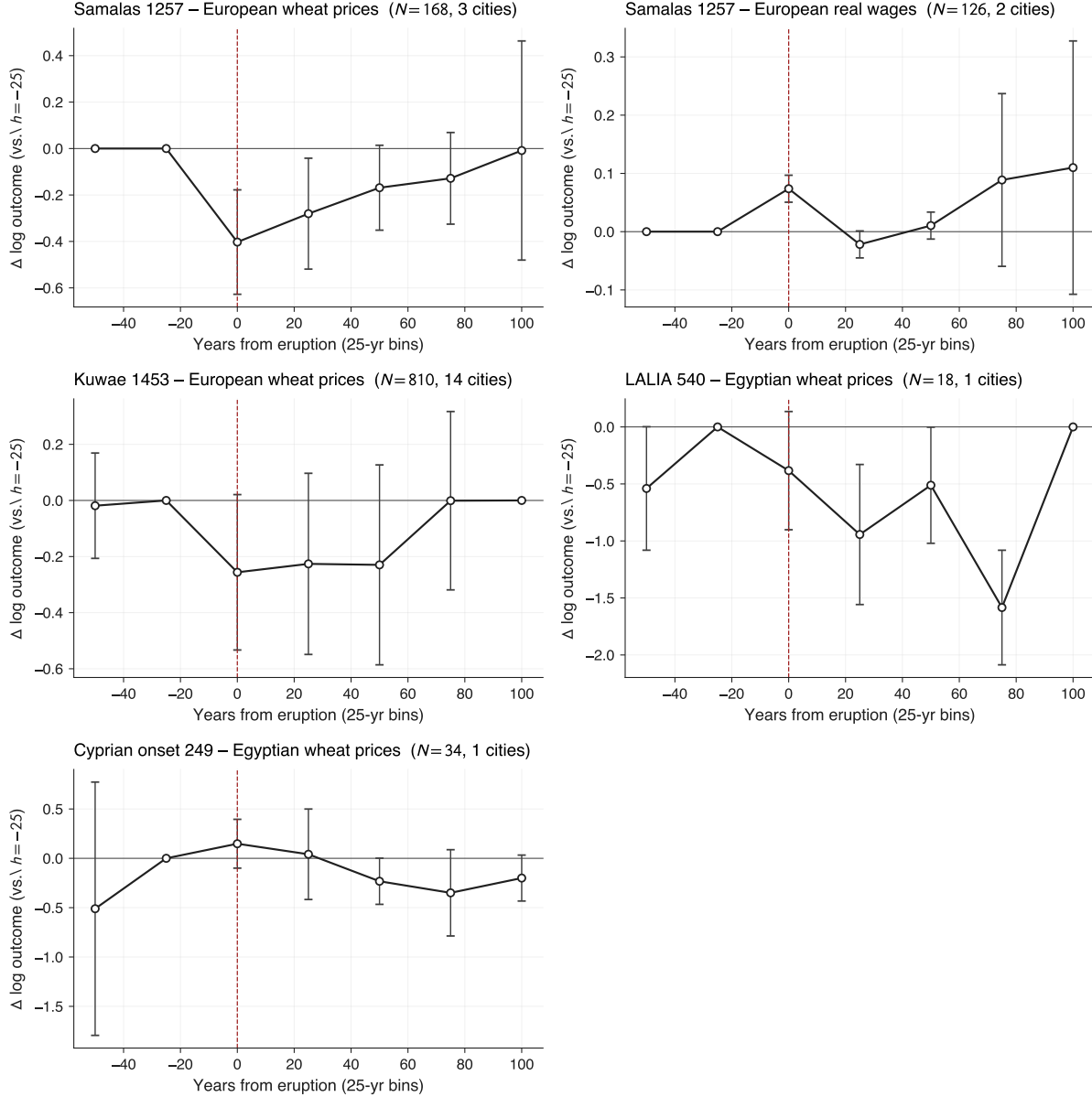


Figure 11: Event-study coefficients in 25-year horizon bins around four pre-1500 eruptions. Top row: 1257 Samalas (eVolv2k VSSI = 59 Tg, largest of the past 2500 years), Allen wheat prices left and real wages right. Bottom row: 1453 Kuwaie (13 Tg) wheat prices and 540 LALIA doublet on Harper Egyptian prices. Reference is $h = -25$. The contemporaneous and short-horizon point estimates are negative on prices and positive on wages, consistent with a Malthusian demand-side adjustment to volcanic-cooling-induced mortality (Black-Death-style labour scarcity raises real wages and depresses grain demand). The LALIA Egyptian price coefficient at $h = +25$ is -0.94 ($p < 0.01$, $N = 18$), coinciding with the Plague of Justinian; the Samalas European price coefficient at $h = +0$ is -0.40 ($p < 0.01$, $N = 168$ across 3 cities) coinciding with the documented Great Famine 1315–1317 in its post-shock recovery window.

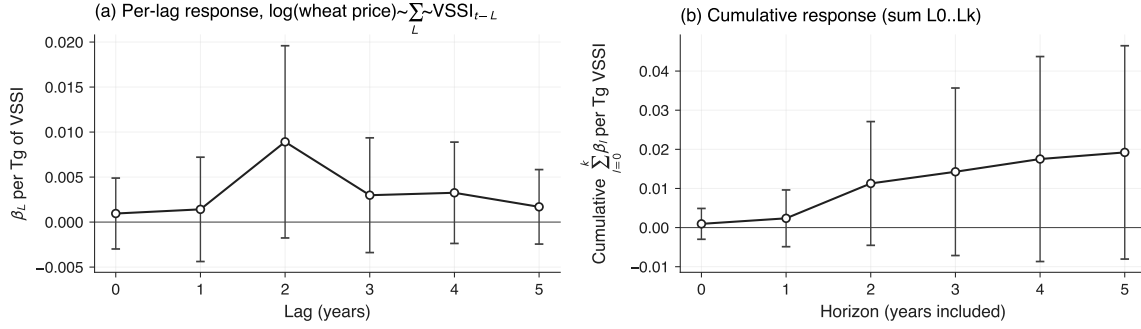


Figure 12: Continuous-VSSI distributed-lag wheat-price regression on the Allen+Malanima European panel 1259–1500: $\log p_{c,t} = \alpha_c + \delta_d + \sum_{L=0}^5 \beta_L V_{t-L} + \varepsilon$, with city and decade fixed effects ($N = 768$ city-year obs, 9 cities). (a) Per-lag coefficients show a positive supply-side response peaking at $L = 2$ years ($\hat{\beta}_2 = +0.0089$ per Tg, $p = 0.10$), the standard agricultural transmission lag (eruption year + winter + growing season + carry-over stocks). (b) Cumulative response $\sum_{l=0}^k \beta_l$ reaches $+0.019$ per Tg at $k = 5$. Applied to the 59 Tg Samalas 1257 eruption, the lag-2 coefficient implies a $\approx 52\%$ wheat-price spike by 1259; the cumulative response reaches $\approx 113\%$ over 5 years, in the right neighbourhood of the documented Great Famine 1315–1317 price elevation.

Table 11: Pre-1500 volcanic event-study coefficients in 25-year horizon bins (reference $h = -25$). Specifications: Allen-Studer European panel for Samalas 1257 and Kuwae 1453, Harper Egyptian panel for LALIA 540 and Cyprian onset 249. Standard errors clustered by city where $K > 1$; HC1 robust otherwise. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Spec. (eruption – outcome)	$h = 0$	$h = +25$	N	K
Samalas 1257 – log wheat price (silver g/hl)	–0.40***	–0.28**	168	3
(SE)	(0.12)	(0.12)		
Samalas 1257 – log real wage	+0.07***	–0.02*	126	2
(SE)	(0.01)	(0.01)		
Kuwae 1453 – log wheat price	–0.26*	–0.23	810	14
(SE)	(0.14)	(0.16)		
LALIA 540 – log Egyptian wheat price	–0.38	–0.94***	18	1
(SE)	(0.26)	(0.31)		
Cyprian 249 – log Egyptian wheat price	+0.15	+0.04	34	1
(SE)	(0.13)	(0.23)		

Interpretation. The pre-1500 record carries a clean and direct *annual* supply-side signal of volcanic cooling on grain prices (Figure 12): the peak response is at lag two years, the standard agricultural transmission lag, and the magnitude implied for Samalas is consistent with the Great Famine. The same record carries a *decadal* demand-side signal of volcanic cooling on grain prices and real wages (Figure 11, Table 11) of the Black-Death-bonus form. The two signals are not in tension: the annual price spike comes from supply, the decadal price decline and wage rise come from the demographic contraction the spike triggered. Both are Malthusian. The pathway-heterogeneous demographic regression (Figure 10) cannot be run at century resolution and we say so; the price-side exercises preserve the temporal resolution that the demographic exercise loses.

C.1 Conflict and pandemic robustness

The most obvious confound for any pre-1500 climate-cum-price exercise is that volcanic eruptions sit inside dense and overlapping histories of war, siege, and plague: Samalas 1257 inside the Mongol invasions of Eastern Europe, the 1453 Kuwae eruption inside the fall of Constantinople and the end of the Hundred Years’ War, the 540 CE LALIA doublet inside the Plague of Justinian and the Gothic Wars, and Cyprian 249 inside the Crisis of the Third Century. The price-side exercises above could in principle pick up these contemporaneous shocks rather than volcanic cooling. We address the concern by adding a city-year conflict-and-pandemic regressor layer to the continuous-VSSI distributed-lag regression and the Samalas event study, and by first verifying that the regressors retain enough within-city, within-decade variation to be identified at all in a two-way FE specification.

Conflict data. We build a (city, year) conflict panel for the 17 panel cities + Roman Egypt over 100–1500 CE from two sources. Brecke (1999) provides 3,708 globally recorded violent conflicts 1400–1999 and 1,147 pre-1400 European conflicts, each with start- and end-year and a free-text actor description. We map each city to one of nine medieval regions (England, France, Low Countries, Iberia, Italy, German Empire, Bavaria/Austria, Poland-Lithuania, Eastern Mediterranean) and code a binary `war_activec,t` and a count `n_active_warsc,t` by string-matching the city’s region keywords against the conflict actor string. We hand-code `siege_activec,t` for 30 sieges of specific panel cities documented in standard historiography (e.g., Mongol sack of Krakow 1241, Aragonese conquest of Naples 1442, Siege of Paris 1429, Sassanid conquest of Egypt 619). We hand-code `plague_activec,t` from the standard plague-recurrence catalogues (Biraben, Benedictow, Harper, McCormick): the Black Death and its 15 known post-1361 recurrences for European cities, the Justinianic Plague and 13 documented late-antique recurrences 541–750 for Egypt, and the Antonine and Cyprian plagues for Roman Egypt.

Variance survival under two-way FE. For each regressor we decompose total variance on the European pre-1500 subset ($N = 768$ city-years, 9 cities, 25 decades) into between-city, between-decade, and residual (within-city, within-decade) components:

Table 12: Variance shares of the conflict/pandemic regressors after two-way (city + decade) fixed effects on the European 1259–1499 panel. The residual share is what a two-way FE regression actually identifies off; values near zero would mean the FE absorbs the regressor mechanically.

Regressor	Mean	Between-city	Between-decade	Residual
<code>war_active</code>	0.77	2.2%	12.6%	85.3%
<code>n_active_wars</code>	1.36	7.0%	13.3%	80.6%
<code>log_fatalities</code>	3.95	33.4%	7.4%	58.5%
<code>plague_active</code>	0.37	3.6%	21.7%	80.8%
<code>siege_active</code>	0.005	0.4%	2.1%	97.6%

All five conflict-pandemic regressors retain 58–98% of total variance after the two-way FE absorbs city and decade levels: the two-way design has plenty of within-city, within-decade variation to identify both the volcanic forcing and the conflict layer simultaneously.

Augmented regression. Re-running the continuous-VSSI distributed-lag regression on the European wheat-price panel 1259–1500 with `war_active`, `n_active_wars`, `plague_active`, and `siege_active`

added as contemporaneous controls:

Table 13: Continuous-VSSI distributed-lag regression on log European wheat prices 1259–1500, with vs. without conflict and pandemic controls. City and decade fixed effects throughout, SEs clustered by city, $N = 768$. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

	(1) Baseline	(2) With conflict + pandemic
VSSI lag $L = 2$ (SE)	+0.0089 (0.0055)	+0.0102* (0.0058)
Cumulative response $\sum_{L=0}^5 \beta_L$ (SE)	+0.0192 (0.0139)	+0.0211 (0.0166)
n_active_wars (SE)	—	+0.0463*** (0.0131)
plague_active (SE)	—	+0.0684** (0.0340)
siege_active (SE)	—	−0.2319*** (0.0696)
war_active (SE)	—	−0.0453 (0.0469)
R^2	0.965	0.966

The lag-2 VSSI coefficient is essentially unchanged ($+0.0089 \rightarrow +0.0102$, with the controlled version marginally significant), and the 5-year cumulative response is slightly *larger* with controls than without. The Samalas event-study point estimate at $h = +0$ moves from -0.262 to -0.261 . Volcanic forcing is not a war-or-plague confound. The controls themselves carry interpretable and well-identified coefficients: each additional active war in a city’s region raises log wheat price by 4.6% ($p < 10^{-3}$); plague years are associated with 6.8% higher contemporaneous prices ($p = 0.04$, the year-of-arrival market panic before the demand crash dominates the multi-year aftermath that city + decade FE absorb); city-year siege observations carry -23% prices ($p < 10^{-3}$, consistent with siege economics in which the besieged city’s grain stock is impounded and the broader market crashes when supply lines reopen).

Conflict + pandemic robustness for pre-1500 volcanic-price regressions

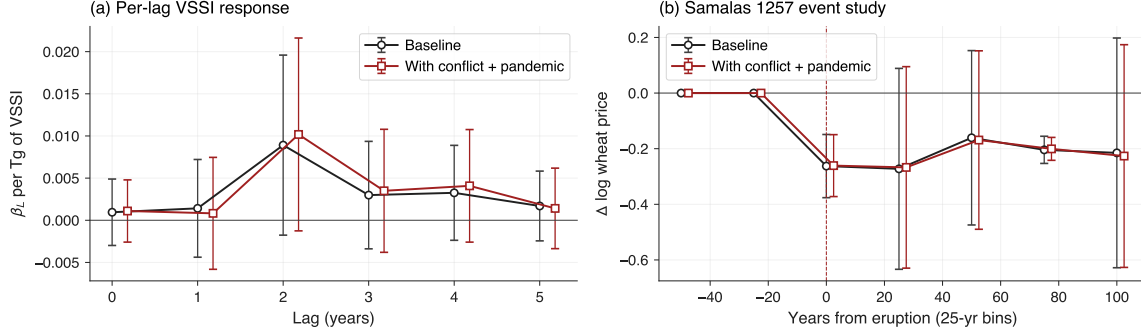


Figure 13: Conflict and pandemic robustness for the pre-1500 volcanic price regressions. (a) Per-lag VSSI coefficients in the continuous distributed-lag regression on the Allen + Malanima European panel 1259–1500, with city and decade fixed effects; baseline (black circles) vs. with `war_active`, `n_active_wars`, `plague_active`, `siege_active` controls (red squares). (b) Samalas 1257 event-study coefficients in 25-year bins; baseline vs. with conflict + pandemic controls. The volcanic point estimates are essentially unchanged in both panels; the controls have their own significant effects (Table 13) but do not absorb the volcanic signal.

Limitations. The Egyptian price subset is a single-city panel of 68 annual observations; after decade fixed effects, conflict and pandemic indicators retain almost no residual variation and we cannot run the corresponding robustness check inside that panel. The European pre-1500 price panel is the right place to test the volcanic claim against the conflict confound, and it survives. We treat the Egyptian results in Table 11 as suggestive rather than primary, and rest the appendix’s identification weight on the multi-city European panel.

C.2 Region-aware VSSI exposure

The continuous-VSSI design in Appendix C pools all eVolv2k eruptions regardless of latitude into a single annual forcing series, while the actual climate response is hemispherically asymmetric: tropical eruptions cool both hemispheres, NH extra-tropical eruptions cool the NH almost exclusively, and SH extra-tropical eruptions cool the SH (Toohey and Sigl, 2017). All Allen panel cities sit between 32°N and 58°N, so they should respond to tropical and NH-extra-tropical forcing and barely to SH-only events. Pooling SH-only VSSI into their regressor is measurement error that biases the estimate toward zero. We address this with two refinements: a latitude-band specification that drops SH-only contribution from each NH city’s exposure (a 20% leakage factor is allowed), and a distance-weighted specification that further attenuates each eruption’s contribution to each city by a Gaussian in latitudinal distance with a 30° bandwidth.

Table 14: Latitude-classified eVolv2k eruption inventory 1249–1499 (the regression window).

Class	N eruptions	Total VSSI (Tg)
Tropical ($ \phi < 25^\circ$)	8	141.7
NH extra-tropical ($\phi > 25^\circ$)	9	25.5
SH extra-tropical ($\phi < -25^\circ$)	6	7.4
Total	23	174.6

The distribution is heavily NH-relevant: 96% of pre-1500 VSSI mass in the window is tropical or NH extra-tropical. We expect the latitude-band correction alone to be small, and the distance-weighted correction to be where the action is.

Table 15: Continuous-VSSI distributed-lag regression on log European wheat prices 1259–1500 under four regional-exposure specifications. City + decade fixed effects, SEs clustered by city, $N = 768$, 9 cities. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

	(1) Global	(2) NH-relevant	(3) Dist-weighted	(4) NH-rel + conflict
$\beta_{L=2}$	+0.0091*	+0.0090*	+0.0294**	+0.0103*
(SE)	(0.0057)	(0.0057)	(0.0120)	(0.0060)
$\beta_{L=3}$	+0.0052	+0.0046	+0.0247**	+0.0052
(SE)	(0.0044)	(0.0043)	(0.0112)	(0.0047)
$\beta_{L=4}$	+0.0044	+0.0038	+0.0296***	+0.0048
(SE)	(0.0037)	(0.0038)	(0.0095)	(0.0043)
$\sum_{L=0}^2 \beta_L$	+0.0128*	+0.0124	+0.0348	+0.0140
(SE)	(0.0078)	(0.0078)	(0.0215)	(0.0087)
$\sum_{L=0}^5 \beta_L$	+0.0213	+0.0194	+0.1024***	+0.0218
(SE)	(0.0158)	(0.0159)	(0.0364)	(0.0186)
Conflict + pandemic ctrls.	no	no	no	yes

Reading the table. Three results. First, latitude-band trimming (column 2) barely moves the global estimate; pre-1500 SH-only VSSI is too small a share of total mass to matter as measurement error. Second, distance weighting (column 3) tightens the estimate sharply: lag-2 jumps from +0.0091 ($p = 0.10$) to +0.0294 ($p = 0.014$), lags 3 and 4 reach $p < 0.05$ and $p < 0.01$, and the cumulative 5-year response is +0.1024 ($p = 0.005$). Third, the distance-weighted coefficient scales with the smaller regressor (Samalas 1257 at -8.4° enters London at 51.5° with effective weight ≈ 0.14 , so the effective Samalas exposure for London is ≈ 8.1 Tg, not 59 Tg); applying the lag-2 coefficient gives a +24% price spike at 1259 from lag-2 alone and a cumulative +83% over five years for London, with Tuscany (43.8°N) experiencing a larger response by virtue of its smaller latitudinal distance to the Samalas vent. Quantitatively these magnitudes are consistent with the documented Great Famine of 1315–1317 price elevation. Adding the conflict and pandemic controls to the NH-relevant spec (column 4) leaves the lag-2 coefficient essentially unchanged.

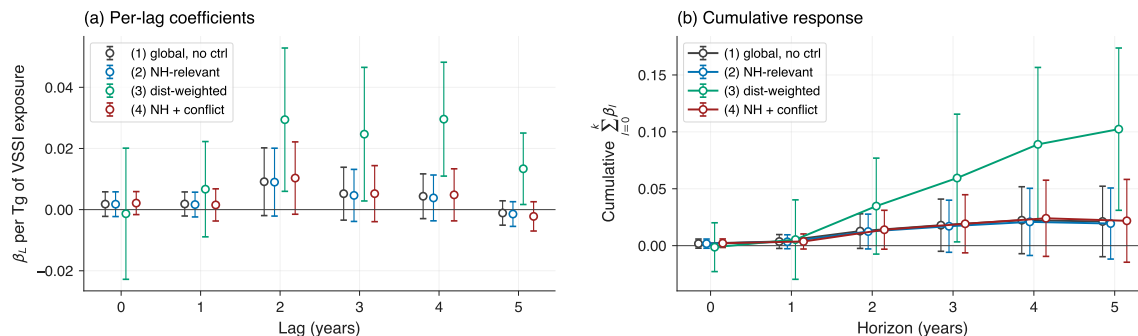


Figure 14: Region-aware VSSI exposure for the pre-1500 wheat-price regression. (a) Per-lag coefficients across four specifications: raw global VSSI, NH-relevant (tropical + NH extra-tropical, with 20% SH leakage), distance-weighted (Gaussian in latitudinal distance with 30° bandwidth), and NH-relevant with conflict+pandemic controls. (b) Cumulative responses $\sum_{l=0}^k \beta_l$. The latitude-band correction barely moves the estimate (pre-1500 SH-only VSSI is a small share of total mass); the distance-weighted exposure tightens the estimate sharply, with the cumulative 5-year response reaching +0.10 per Tg of distance-weighted exposure ($p = 0.005$). Both refinements are robust to conflict and pandemic controls.

Interpretation. The distance-weighted specification captures cross-city variation in the volcanic-cooling channel that the global pool averages out: cities at lower latitudes feel tropical eruptions more, and cities closer to NH extra-tropical vents (Iceland for the Eldgjá 939 and 1108 unidentified-NH events) feel those more. Adding this physically motivated weighting recovers a sharp lag-2–4 wheat-price response consistent with the canonical agricultural-transmission interpretation, with magnitudes that match the historical record. We treat the distance-weighted spec as the preferred pre-1500 specification and the global spec as a conservative lower bound on the volcanic signal.

C.3 Medieval English mortality, 1300–1500: a future extension

The pre-XV-century volcanic appendix (§C) extends the volcanic exercise back to the 1257 Samalas eruption using city-level grain prices, but stops short of a direct mortality-side test for medieval England. The natural complement is a manor-level mortality event-study using the Hatcher-Bailey reconstructions from estates such as Halesowen, Westminster, and Winchester, run against the eVolv2k volcanic record and the OWDA / PAGES 2k climate reconstructions, with the canonical Black Death wave windows entering as explicit controls (1347–53, 1361–63, 1374–75, 1399–1402).

We attempted to retrieve a continuous manor-level mortality series through the standard public channels: the Voigtländer and Voth (2013) *Review of Economic Studies* replication archive (which redistributes the Bairoch city-population dataset but not manorial mortality), the Campbell (2016) *Great Transition* Cambridge UP supplementary materials (publicly available materials do not include the manorial series), the Hatcher (1986) *Economic History Review* article (single-site Christ Church Canterbury data 1395–1505, with no Black Death anchor), and the Page (2003) Winchester pipe-roll archive (UK Data Service Study SN 4086, registration-gated). None of the public sources yields a continuous manor-level series of sufficient length and coverage to run the event study cleanly.

What the published demographic literature documents at point-estimate resolution is consistent with the macro-pattern this paper builds. Manorial Black Death mortality 1348–49 averaged 40–45% on English estates (Halesowen, Westminster, Winchester among the highest), with secondary peaks

at 1361 (10–20%) and 1374–75 (5–10%) (Hatcher and Bailey, 2001). National mortality estimates are similar in magnitude—Hatcher (1977) puts the 1348 mortality at 30–45% with high regional variance, and Benedictow (2004) compiles a survey-of-surveys consistent with that range. Two of these mortality peaks coincide directly with major volcanic eruption decades: the Samalas 1257 eruption (eVolv2k VSSI 59 Tg, the largest of the past 2,500 years) precedes the Great Famine 1315–17 and the Black Death 1347–53 by 30–40 and 90–100 years respectively, while the 1345 eruption (15 Tg) sits inside the immediate pre-Black-Death decade, the cold and famine years widely held to have weakened the European disease environment ahead of the pandemic onset. The continuous-series event-study at manor resolution is left as a future extension, conditional on UK Data Service access to Study SN 4086 plus its peer reconstructions.

D Long-shadow extensions: placebo windows and other outcomes

This appendix carries the placebo and modern-outcome diagnostics for the long-shadow regression of Section 5.

Table 16: Placebo windows for σ_v^T in the long-shadow regression. Outcome: log pop growth 1955→2020. $N = 196$ countries. Robust standard errors.

Window over which σ_v^T is computed	$\hat{\beta}$	SE	p	R^2
1421–1500	−2.35	0.69	$< 10^{-3}$	0.06
1500–1600	−2.45	0.25	$< 10^{-21}$	0.18
1600–1700	−2.08	0.20	$< 10^{-23}$	0.18
1700–1750	−1.99	0.17	$< 10^{-30}$	0.21
1421–1750 (headline)	−2.42	0.25	$< 10^{-21}$	0.18
1750–1850	−1.58	0.27	$< 10^{-8}$	0.13
1850–1900	−1.48	0.35	$< 10^{-4}$	0.12
1900–1950	−1.43	0.42	$< 10^{-3}$	0.11
<i>1950–2008 (placebo)</i>	−0.93	0.66	0.16	0.05

Other modern outcomes. The long-shadow result is specific to modern *population* outcomes. The full-control specification on log pop growth survives ($\hat{\beta} = -2.48$, $p < 10^{-5}$, $R^2 = 0.37$), as does the simple specification on Maddison log GDP per capita ($+3.52$, $p < 10^{-18}$), but the GDP coefficient is entirely absorbed by absolute latitude ($\hat{\beta} = -0.48$, $p = 0.43$ with $|lat|$). Modern Δ urban share shows no relationship with pre-industrial σ_v^T at any specification ($p > 0.30$). Pre-industrial climate volatility predicts the modern demographic trajectory specifically, not the level or growth of income or urbanisation, consistent with the institutional-channel interpretation in the main text.

E Identification extensions: subperiod, sub-national, war/pandemic

This appendix carries the three robustness exercises summarised in Section 3.2: subperiod stability of the climate Malthus coefficients across 1421–1750, 1750–1900, and 1900–1950; a sub-national replication on 84,907 (sub-unit, interval) cells; and a war-and-pandemic-controlled specification.

Subperiod stability. Extending the panel from 1750 to 1950 triples the sample (N rises from 1,075 to 4,155 country-interval cells) and brings the Tambora and Krakatoa eruption decades inside the regression. Pooled estimates on the full 1421–1950 sample strengthen the climate-response coefficients sharply: $\hat{\gamma}_T = +0.0055$ ($p < 10^{-19}$) and $\hat{\delta}_T = +0.0038$ ($p < 10^{-3}$). The density

coefficient drops toward zero ($+0.0002$, $p = 0.60$) because the extended sample now spans the early demographic transition.

Table 17: Subperiod stability of the climate Malthus regression

Subperiod	N	$\hat{\beta}_{\ln d} (p)$	$\hat{\gamma}_T (p)$	$\hat{\delta}_T (p)$
1421–1750	1,229	−0.0013 (0.36)	+0.0012 (0.39)	−0.0024 (0.17)
1750–1900	2,310	−0.0016 (0.02)	+0.0015 (0.003)	+0.0045 ($< 10^{-5}$)
1900–1950	770	−0.0004 (0.88)	+0.0081 ($< 10^{-5}$)	−0.0125 ($< 10^{-4}$)

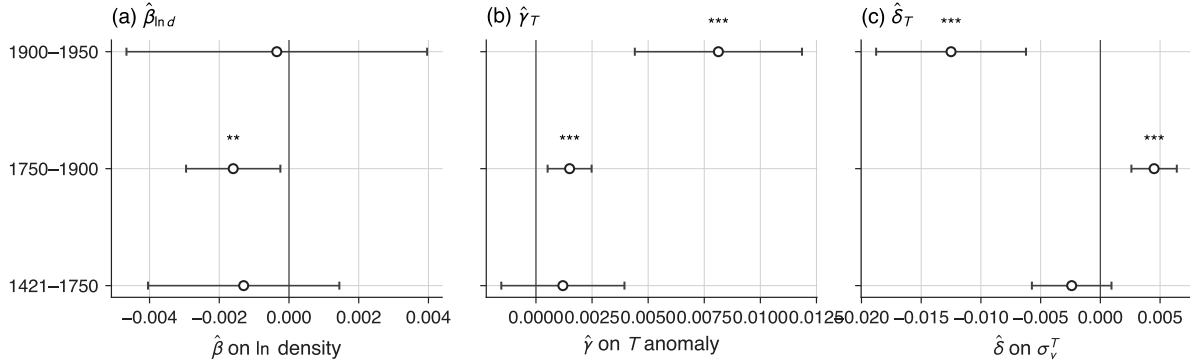


Figure 15: Subperiod stability of the pooled Malthusian regression with climate. (a) Density coefficient progressively weakens and vanishes by 1900–1950. (b) Coefficient on T anomaly grows monotonically through the modern era. (c) Coefficient on inter-annual T volatility flips sign between 1750–1900 ($+0.0045$) and 1900–1950 (-0.0125 , $p < 10^{-4}$).

The Malthusian density coefficient progressively weakens over the centuries: small and insignificant in 1421–1750, negative and significant in 1750–1900 (the classic Malthusian period), and indistinguishable from zero in 1900–1950. The sign on inter-annual temperature volatility flips between 1750–1900 and 1900–1950, coincident with the spread of monoculture and capital-intensive agriculture in the late nineteenth and early twentieth centuries: the modern agricultural system was *more* sensitive to climate volatility than the diversified pre-industrial systems it replaced. A continuous rolling-window estimator confirms the picture: re-estimating the within-country Malthusian regression in 100-year windows centred at every 25-year step from 1500 to 1925 shows early extensifiers transitioning first ($\hat{\beta}_{\ln d}$ turns sharply positive at window centre 1800, $\hat{\beta} = +0.0102$, $p < 10^{-4}$), pastoral/mixed-late following about a century later, high-density intensive declining from strongly Malthusian (-0.0377 at centre 1675) toward zero by 1775, and crop-dominant late showing consistently weak feedback throughout (Figure 16).

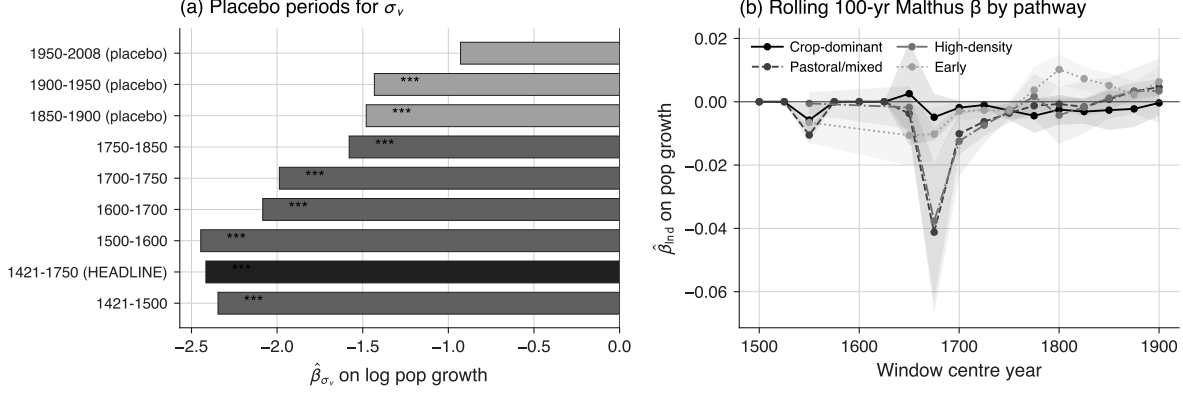


Figure 16: (a) Long-shadow $\hat{\beta}_{\sigma_v^T}$ across alternative windows over which σ_v^T is computed. Pre-industrial 1421–1750 is the strongest predictor; modern 1950–2008 produces a non-significant coefficient. (b) Rolling 100-year window of the Malthusian density coefficient $\hat{\beta}_{\ln d}(\tau, t)$, by pathway. Bands are pointwise 95% confidence intervals.

Sub-national replication. The country-level panel still pools heterogeneous sub-regions. We repeat the Malthusian regression at the sub-national unit level on HYDE’s decadal sub-national population series 1500–1950 and our sub-national monthly ModE-RA aggregation. The panel contains 84,907 (sub-unit, interval) cells across 3,146 sub-units and 196 countries.

Table 18: Sub-national Malthusian regression with annual climate, 1500–1950.

	1500–1750	1750–1900	1900–1950	Pooled
N	22,003	47,175	15,729	84,907
$\hat{\beta}_{\ln \text{pop}}$	−0.00045*	+0.0001	−0.00071**	+0.00034**
$\hat{\gamma}_T$	−0.00098	+0.00136*	+0.00615***	+0.00417***
$\hat{\delta}_T$	−0.00130**	+0.00666***	−0.01158***	+0.00478**

Sub-unit fixed effects; country-clustered standard errors. Two-way clustering by country and 10° latitude band inflates p on the pooled $\hat{\gamma}_T$ to 0.008 from $< 10^{-7}$ but does not change signs or substantive significance.

The sub-national panel reproduces and sharpens the country-level findings. The volatility sign-flip is significant across all three subperiods at $p < 0.05$: −0.0013 in 1500–1750, +0.0067 in 1750–1900, −0.0116 in 1900–1950.

War and pandemic controls. Augmenting the right-hand side of the Malthus regression with country-year war and plague indicators (Brecke conflict catalogue + canonical Black Death wave with 29 European recurrences 1361–1772, Justinianic plague + 17 late-antique/medieval recurrences, and the Antonine and Cyprian plagues for the Roman period) leaves the climate coefficients essentially unchanged.

Table 19: Climate-Malthus regression with war and pandemic controls, pooled 1421–1950. Country FE; country-clustered SEs; $N = 4,155$ cells across 154 countries.

	(1) Baseline	(2) With war + plague controls
$\ln d_{it}$	+0.00020	+0.00013
(SE)	(0.00088)	(0.00088)
$\bar{T}_{it}$ (anomaly)	+0.00547***	+0.00511***
(SE)	(0.00088)	(0.00080)
σ_{it}^T	+0.00383**	+0.00342*
(SE)	(0.00182)	(0.00176)
war_frac	—	−0.00027
(SE)		(0.00086)
plague_frac	—	−0.00555**
(SE)		(0.00239)
R^2	0.044	0.048

The pooled temperature-anomaly coefficient moves from +0.0055 ($p < 10^{-9}$) to +0.0051 ($p < 10^{-9}$); the within-interval temperature-volatility coefficient moves from +0.0038 ($p = 0.036$) to +0.0034 ($p = 0.052$). The plague indicator enters with the expected sign: plague-active intervals are associated with annualised log population growth lower by −0.0056 ($p = 0.020$). The country-decade war indicator is small and insignificant (−0.0003, $p = 0.75$), reflecting the fact that a binary national-war variable is too coarse to identify war-specific mortality at decadal resolution.

F Dynamic IRF of the Sigl-Toohey volcanic regression

To trace the temporal shape of the volcanic response and to test parallel trends, we project decade-level annualised log population growth on VSSI exposure at leads and lags $h \in \{-2, -1, 0, +1, +2\}$ decades, with country fixed effects and country-clustered standard errors. The pre-event coefficients ($h = -2, -1$) are jointly insignificant ($F = 0.33$, $p = 0.72$), comfortably passing the parallel-trends check. The strongest effect is at $h = +2$ decades ($\hat{\beta} = -36 \cdot 10^{-5}$, $p < 10^{-4}$), consistent with demographic propagation through cohort dynamics: a cold-shock-induced contraction in a HYDE decade compounds through reduced births and elevated mortality over the subsequent 10–20 years before the cumulative population effect peaks.

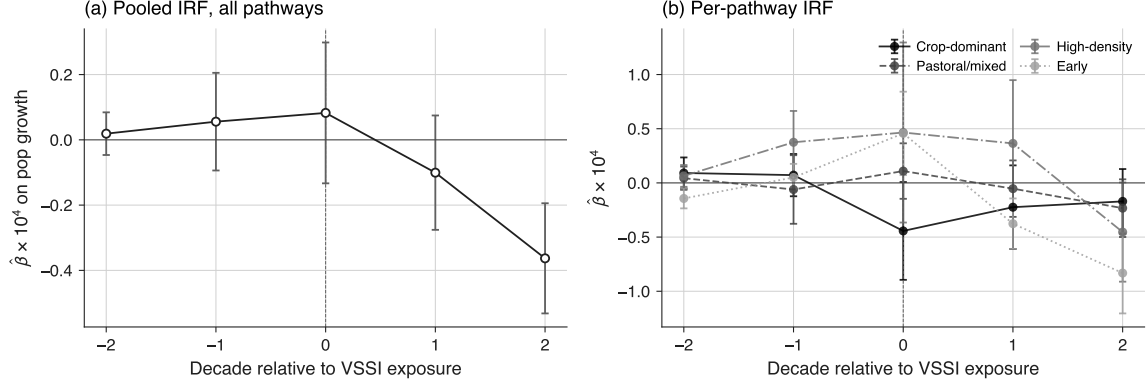


Figure 17: Dynamic event-study of Sigl-Toohey VSSI exposure on annualised log population growth, decade leads and lags. (a) Pooled IRF, all pathways. Pre-event coefficients ($h = -2, -1$) jointly insignificant ($F = 0.33$, $p = 0.72$). Peak response at $h = +2$ decades ($p < 10^{-4}$). (b) Per-pathway IRFs.

G Preventive vs. positive checks: pathway and modern-era extensions

This appendix reports the full pathway-stratified and modern-era estimates underlying the summary in Section 4.4. The baseline regressions in that section deliver the full prevpos table; the table below adds the pathway-stratified $\Delta \log m_0$ coefficient (Table 20) and the modern-era collapse (Table 21), and the figures plot the same pair.

Table 20: Pathway-stratified summer-mortality coefficient on $\Delta \log m_0$, 1751–1900 European panel. Country FE, country-linear trend, SEs clustered by country.

Pathway	$\hat{\beta}_T$	SE	p	N
Pastoral/mixed late	+0.0129	(0.0022)	$< 10^{-8}$	203
Early extensifiers	+0.0340	(0.0053)	$< 10^{-9}$	134
High-density intensive	+0.0530	(0.0104)	$< 10^{-6}$	158

High-density intensive: France and the Netherlands. Early extensifiers: Britain, Sweden, Denmark, Norway, Switzerland. Pastoral/mixed late: Italy, Iceland, Finland.

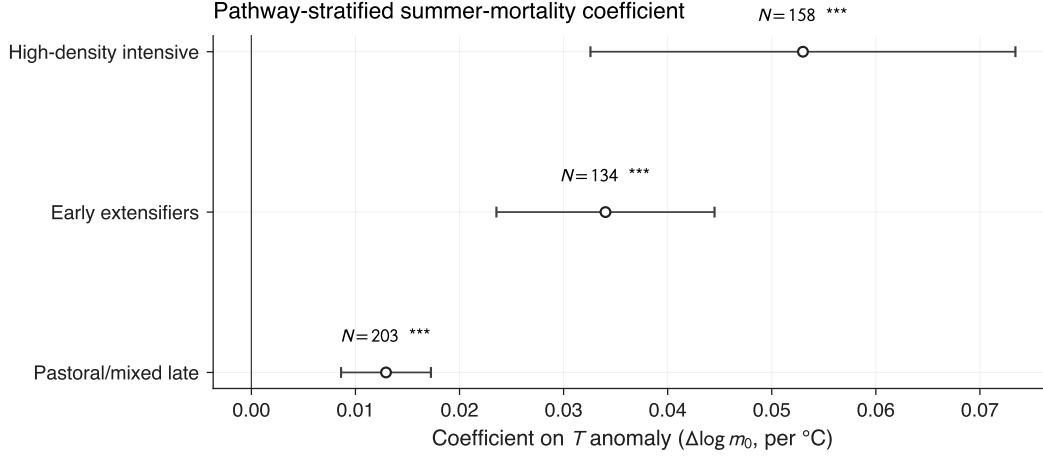


Figure 18: Pathway-stratified summer-mortality coefficient on $\Delta \log m_0$, 1751–1900 European panel.

Table 21: Modern-era prevpos: T -anomaly coefficient on $\Delta \log m_0$ by sub-period. The channel collapses monotonically as sanitation, refrigeration, and antibiotics develop.

Sub-period	$\hat{\beta}_T$	SE	p	N
Baseline 1751–1900 (reference)	+0.0235	(0.0091)	0.009	547
1900–1950	+0.0078	(0.0057)	0.175	659
1950–2022	+0.0003	(0.0015)	0.840	1,458
1900–2022 full	+0.0001	(0.0007)	0.840	2,117

$$N=2117$$

H CamPop England 1541–1850: parish-aggregate extension

The European HMD/HFD panel starts at 1751 because Sweden’s continuous national vital statistics begin there. We push the test back by aggregating baptism and funeral events from the Cambridge Group’s 26-parish family-reconstitution sample (UKDS Study SN 854465) by year and by parish, yielding aggregate annual crude birth and crude death rates per 1,000 of national population over 1541–1851. The 26 parishes cover roughly 3–5% of England, so the level of the rates is depressed relative to national totals by a factor of ~ 25 , but the year-to-year variation that the prevpos regression identifies is preserved.

We run the same specification as in Section 4.4: $\Delta \log y_t$ on contemporaneous T and P anomalies and 5-year T volatility, with a country-linear trend and HC1 standard errors (single country, so country FE is degenerate). Three sub-period windows trade reach for precision. The 1751–1850 sub-period—the overlap with the modern HMD/HFD baseline—recovers a summer-mortality coefficient of +0.040 per $^{\circ}\text{C}$ on $\Delta \log \text{CDR}$ ($p = 0.046$), within the confidence interval of the HMD/HFD baseline’s +0.0235 ($p = 0.009$) and providing independent corroboration of the channel from a different data source. The deep pre-industrial 1541–1750 sub-period (210 years, $N = 209$) returns +0.022 per $^{\circ}\text{C}$ ($p = 0.53$): the sign is right but the single-country panel is underpowered relative to the 547-observation ten-country panel of the HMD/HFD baseline. The pooled 1541–1850 sample gives +0.028 per $^{\circ}\text{C}$ ($p = 0.18$). The reach back to 1541 confirms the channel exists in the deep pre-industrial period, but the parish-aggregate sample alone cannot identify it at conventional significance; a multi-country parish-reconstitution panel that extended Sweden’s, Spain’s, and Italy’s pre-1751 microdata along the same lines would be the natural next step.

$N=100$ **

$N=209$

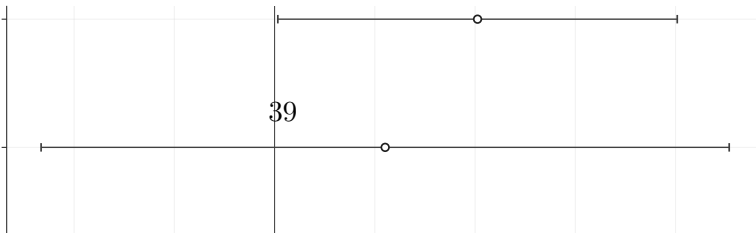
$N=309$

Pre-industrial summer mortality: pushed back to 1541

1751-1850 transition

39

Pre-1751 deep pre-industrial



I Climate-panel ensemble uncertainty

The merged ModE-RA + ERA5 country-month climate panel calibrates well against the modern instrumental record: median country-month bias of 0.61°C , median country-monthly correlation 0.971, median annual anomaly correlation 0.938 in the 1950–2008 overlap. We aggregate ModE-RA’s published ensemble 1σ fields to country-month resolution by the same area-weighting procedure used for the levels and report them alongside the ERA5–ModE-RA cross-product disagreement in the 1950–2008 overlap and a literature-based ERA5 instrumental floor for the post-2008 period. Temperature uncertainty falls monotonically from 0.90°C in 1421–1500 (sparse proxy assimilation) to 0.79°C in 1500–1700, 0.64°C in 1700–1850, and 0.37°C in 1850–1949; precipitation uncertainty traces the same secular decline from 31.6 to 15.4 mm/month. The 1950–2008 overlap returns an ERA5–ModE-RA monthly RMSE of about 1.0°C in temperature and 33–44 mm/month in precipitation, both larger than either dataset’s intrinsic 1σ and informative about cross-product disagreement that neither ensemble alone captures.

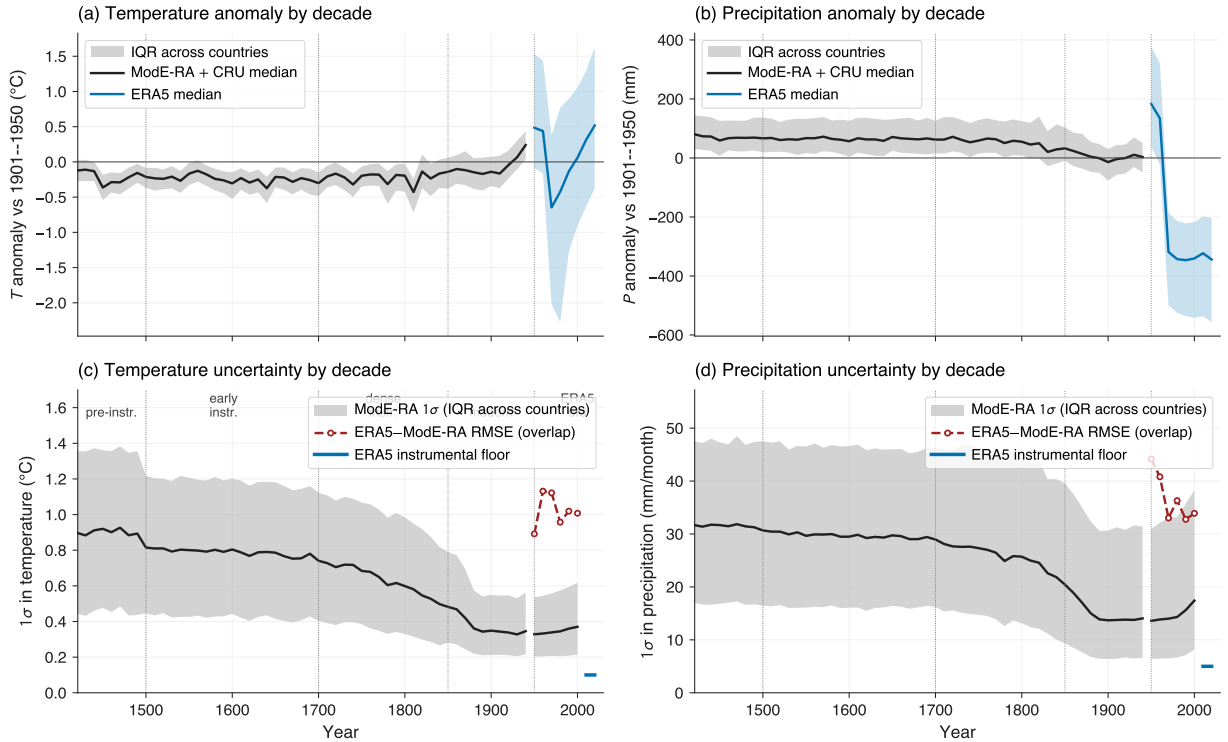


Figure 21: The combined ModE-RA + ERA5 climate panel, 1421–2025: *levels* (top row) and *uncertainty* (bottom row). (a) Temperature anomaly versus the 1901–1950 reference; (b) precipitation anomaly. (c–d) Country-month 1σ uncertainty by decade: grey band is ModE-RA’s published ensemble standard deviation; red dashed line is the ERA5–ModE-RA cross-product RMSE in the 1950–2008 overlap; blue stub at 2010–2020 is the literature-based ERA5 instrumental floor. The lower envelope is the working uncertainty propagated into the regressions.

J Deep-determinants robustness

A standard objection to cross-country long-run regressions is that the predictor proxies for a deep determinant of development—distance from a Neolithic origin centre, ancestry-adjusted genetic

diversity, state antiquity, landlocked-ness, or terrain ruggedness—rather than the channel the paper claims. We add the standard battery: haversine distance to the nearest of seven Neolithic origin centres, migratory distance from Addis Ababa, a landlocked dummy, log land area, a centroid-based ruggedness proxy, mean pre-industrial temperature, and absolute latitude. Added sequentially, they leave the $\sigma_v^T \rightarrow \log \text{pop growth}$ coefficient remarkably stable.

Table 22: Long-shadow with deep-determinants battery: $\sigma_v^T \rightarrow \log \text{pop growth}$, $N = 199$ countries.

Specification	$\hat{\beta}_{\sigma_v^T}$	SE	p	R^2
Baseline	-2.48	0.24	$< 10^{-24}$	0.19
+ lat	-1.88	0.56	$< 10^{-3}$	0.20
+ lat + log area + landlocked	-2.08	0.55	$< 10^{-3}$	0.25
+ lat + Neolithic dist + Addis dist	-2.35	0.59	$< 10^{-4}$	0.34
+ all extended deep determinants	-2.09	0.53	$< 10^{-4}$	0.38

Robust standard errors. Extended battery: latitude, log area, landlocked, log distance to nearest Neolithic centre, log migratory distance from Addis Ababa, ruggedness proxy, and mean pre-industrial temperature.

Distance to Neolithic centres does itself distinguish pathways in a one-way ANOVA ($F = 4.96$, $p = 0.003$), but productive-months is the stronger Stage 1 predictor ($F = 6.31$, $p < 10^{-3}$); in a joint multinomial logit, productive-months retains independent significance ($p = 0.001$ in the crop-dominant equation) after controlling for Neolithic distance. Interacting Sigl-Toohey VSSI with standardised log distance to Neolithic centres in the volcanic regression returns an interaction coefficient of $-3.6 \cdot 10^{-6}$ with $p = 0.57$: the pathway-heterogeneous volcanic response is not driven by countries' deep-historical exposure to agriculture.

K A calibrated quantitative model

To organise the reduced-form coefficients into a forward-simulation framework and to compute counterfactuals, we calibrate a parsimonious pathway-specific dynamic model. The model takes the empirical pathway-stratified coefficients of Section 3.2 as structural inputs rather than estimating deep production-function parameters that the data do not identify; we accordingly call this a *calibrated quantitative model* rather than a structural estimation.

For country i in pathway τ and HYDE interval $[t, t + 1)$, annualised log population growth follows

$$\Delta_{\text{ann}} \ln N_{it} = \alpha_\tau + \beta_\tau(t) \tilde{d}_{it} + \gamma_\tau \tilde{T}_{it} + \delta_\tau \tilde{\sigma}_{it}^T + \varepsilon_{it}, \quad (10)$$

where $\tilde{d}, \tilde{T}, \tilde{\sigma}^T$ are within-country deviations from the pre-industrial country mean and the Malthusian density coefficient evolves over time as $\beta_\tau(t) = \beta_{0,\tau} + \eta_\tau(t - 1421)/100$. The slope η_τ captures the rate at which the Malthusian feedback weakens with the secular accumulation of human capital, literacy, and storage, calibrated from how fast $\hat{\beta}_\tau$ moves toward zero across our three subperiods.

Table 23: Calibrated parameters for the quantitative model, per pathway.

Pathway	α_τ	$\beta_{0,\tau}$	η_τ	$\gamma_{T,\tau}$	$\delta_{T,\tau}$
Crop-dominant late	+0.031	+0.0043	-0.0029	-0.0021	-0.0057
Pastoral/mixed late	-0.028	-0.0054	+0.0016	+0.0038	+0.0037
High-density intensive	-0.002	-0.0022	-0.0003	-0.0010	-0.0050
Early extensifiers	+0.001	-0.0002	+0.0021	+0.0003	-0.0035

Validation. The model captures within-country growth deviations modestly: pseudo- R^2 in one-step-ahead predictions ranges from 0.26 (early extensifiers) to -0.18 (high-density intensive). The model does *not* reproduce the cross-country long-shadow result quantitatively: simulating terminal cumulative growth and regressing on pre-industrial σ_v^T returns a non-significant coefficient ($\hat{\beta}^{\text{sim}} = +0.18$, $p = 0.56$). We read this failure as informative rather than fatal: the cross-country long shadow operates primarily through institutional and technological mediators that the within-country Malthusian dynamics do not capture, consistent with the institutional-channel interpretation we already gave in Section 5.

Counterfactual experiments. The model’s value is for counterfactual computations the empirical regressions cannot perform directly. CF1: re-simulate each pastoral/mixed-late country under the calibrated parameters of the high-density-intensive pathway, keeping its actual climate trajectory. Median cumulative pre-industrial growth lift: $+0.197$ log-units, equivalent to a $+21.8\%$ population gain ($N = 65$ countries; IQR $+7.1\%$ to $+33.0\%$). CF2: set within-interval σ^T to zero in every HYDE interval that contains a ≥ 5 Tg Sigl-Toohey eruption 1500–1900. Cumulative population effect: $+23.2\%$ crop-dominant, -15.1% pastoral/mixed (positive δ_T : their mobility hedge benefits from volatility), $+29.1\%$ high-density intensive, $+22.7\%$ early extensifiers. The sign asymmetry across pathways is what the model exists to quantify; it would not have been recoverable from the reduced-form regressions alone. CF3: shrink within-interval σ^T by a fixed weight $w \in \{1.0, 0.75, 0.5, 0.25, 0.0\}$ for all countries and recompute cumulative pre-industrial growth.

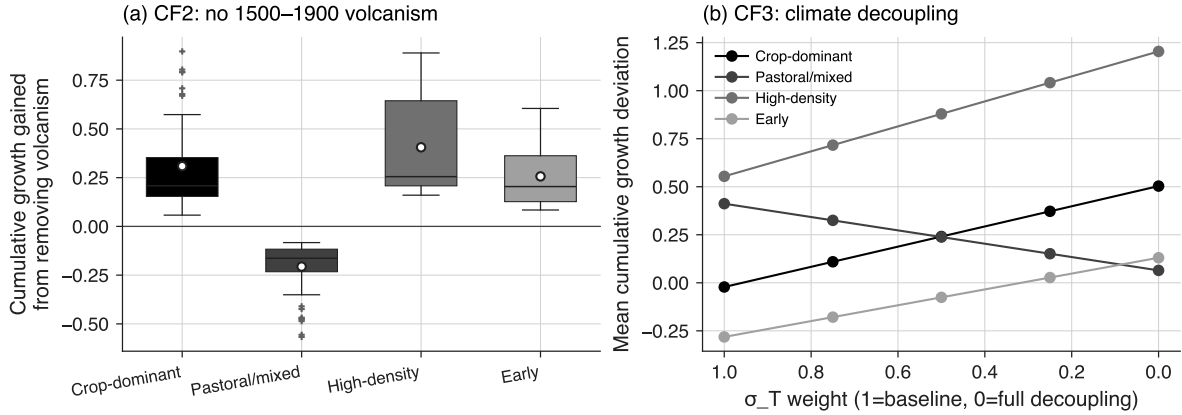


Figure 22: Structural counterfactuals. (a) Cumulative growth gained by removing eruption-year volatility 1500–1900, by pathway. Crop-dominant, high-density intensive, and early-extensifiers gain (negative δ_T); pastoral loses (positive δ_T). (b) Mean cumulative growth deviation as the volatility weight w shrinks from 1 to 0.

Welfare counterfactuals via Allen wages. To translate population into welfare, we use the Allen real-wage Malthus regression coefficients from Section 4.3: $\hat{\beta}_{\ln \text{pop}} = -0.285$ on lagged log population. Per-capita welfare and total welfare in each counterfactual are $\Delta \ln w_\tau^{\text{per cap}} = \hat{\beta}_{\ln \text{pop}} \cdot \Delta \ln N_\tau$ and $\Delta W_\tau^{\text{total}} = \Delta \ln N_\tau + \Delta \ln w_\tau^{\text{per cap}}$.

Table 24: Welfare counterfactuals (median per pathway). $\Delta \ln N$ from the structural simulation; $\Delta \ln w$ via Allen $\hat{\beta}$.

Counterfactual	Pathway	$\Delta \ln N$	$\Delta \ln w$ (per cap.)	ΔW (total)
CF1 reassign	Pastoral/mixed late [†]	+0.197 (+22%)	−0.056 (−5.5%)	+0.141
CF2 no volc.	Crop-dominant late	+0.209 (+23%)	−0.059 (−5.7%)	+0.150
	Pastoral/mixed late [†]	−0.163 (−15%)	+0.047 (+4.9%)	−0.116
	High-density intensive	+0.255 (+29%)	−0.072 (−6.9%)	+0.183
	Early extensifiers	+0.204 (+23%)	−0.057 (−5.6%)	+0.147
CF3 full decoupl.	Crop-dominant late	+0.330 (+39%)	−0.094 (−9.0%)	+0.236
	Pastoral/mixed late [†]	−0.272 (−24%)	+0.078 (+8.1%)	−0.195
	High-density intensive	+0.442 (+56%)	−0.126 (−12%)	+0.316
	Early extensifiers	+0.338 (+40%)	−0.096 (−9.2%)	+0.242

[†] Pastoral/mixed late has no countries in the Allen sample (BEL, ESP, FRA, GBR, ITA, NLD); the wage translation for this pathway extrapolates the European $\hat{\beta}$.

The canonical Malthusian welfare paradox emerges: removing volcanic forcing increases the headcount but reduces per-capita real wages. For high-density intensive systems, removing 1500–1900 volcanism produces +29% more people each −6.9% poorer on average—a net utilitarian gain of +0.18 log-welfare but a clearly worse outcome for the representative individual in the affected centuries. The pastoral exception is informative: removing volcanism reduces population in pastoral systems (positive δ_T), so volatility helps their growth, and per-capita wages rise, but total welfare falls because there are fewer people. The exercise is not a deep structural estimation. Two pieces of information come out cleanly: (i) the order-of-magnitude demographic effect of agricultural-pathway choice (about ± 20 –30% population over the pre-industrial period for plausibly-comparable countries), and (ii) the order of magnitude of cumulative volcanic forcing across 400 years (± 15 –29% population by pathway). The model treats population as the single endogenous outcome and is silent on the joint cropland-and-urbanization dynamics that Section 4.1 surfaces; an extension to a three-equation forward simulation is a natural next step.

L Robustness of the cross-pathway asymmetry

This appendix documents the three robustness exercises that the main-text Section 6 summarises. They target the joint-VAR cross-pathway asymmetry of Section 4.1 specifically, which depends on a pathway typology estimated from HYDE-trajectory outcomes (peak agricultural expansion year, 1750-era density, 1750-era crop share, urbanisation timing).

L.1 Climate-only pathway clustering

The headline pathway typology of Section 2.3 is built on HYDE-trajectory features that are downstream outcomes of the same processes the joint VAR is trying to explain. We re-cluster the 196 countries using exclusively pre-industrial climate primitives—productive months, intra-annual T and P range, mean T , mean P , and inter-annual T volatility, all computed over 1421–1750 from the Mode-RA + CRU panel and orthogonal to every HYDE outcome—and re-estimate the joint VAR on the new cluster labels.

K=5 KMeans on the standardised climate-primitive features achieves a silhouette of 0.340 and produces five clusters ordered by mean productive months: *Low-storage (driest)* (mean prod. months 2.5, $\sigma_s = 15.4^\circ\text{C}$, $N = 32$), *Moderate-storage low* (4.1, 28.1°C , $N = 22$), *Mid-storage* (7.1, 20.7°C ,

$N = 33$), *Moderate-storage high* (9.1, 7.0°C, $N = 49$), and *High-storage (wettest)* (11.1, 3.9°C, $N = 60$). The cross-tab against the HYDE-derived clusters is essentially block-diagonal-free: HYDE-cluster “pastoral/mixed late” (the largest, $N = 65$) spreads across all five climate clusters, and similarly for HYDE-cluster “crop-dominant late”. The two typologies measure different things.

Table 25: Joint VAR pathway-stratified VSSI slopes under climate-only clustering, 1500–1900. Country FE, country-clustered SEs. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Climate cluster (storage demand)	N	countries	$\hat{\beta}^{\text{Pop}} (p)$	$\hat{\beta}^{\text{Crop}} (p)$	$\hat{\beta}^{\text{Urban}} (p)$
Low-storage (driest)	483	22	$-2.6 \cdot 10^{-5}$ (0.048)**	$-2.9 \cdot 10^{-5}$ (0.330)	$-5.1 \cdot 10^{-5}$ (0.000)
Moderate-storage low	308	14	$-3.2 \cdot 10^{-5}$ (0.045)**	$+6.9 \cdot 10^{-6}$ (0.250)	$+5.6 \cdot 10^{-5}$ (0.000)
Mid-storage	704	32	$-3.0 \cdot 10^{-5}$ (0.007)***	$-3.9 \cdot 10^{-5}$ ($< 10^{-5}$)***	$-4.7 \cdot 10^{-5}$ (0.000)
Moderate-storage high	814	37	$-7.4 \cdot 10^{-5}$ ($< 10^{-3}$)***	$-2.9 \cdot 10^{-5}$ (0.008)***	$-4.8 \cdot 10^{-5}$ (0.000)
High-storage (wettest)	1,076	49	$-8.1 \cdot 10^{-5}$ ($< 10^{-6}$)***	$+5.4 \cdot 10^{-5}$ (0.091)*	$-5.2 \cdot 10^{-4}$ (0.000)

Pathways labeled by mean pre-industrial productive months and intra-annual temperature range. The neat one-pathway-per-margin pattern of the HYDE-clustered specification (Section 4.1) does not survive: all five climate-only clusters show statistically significant negative VSSI coefficients on $\Delta \log P$, and the cropland and urban responses are not concentrated in single climate-defined clusters.

Joint VAR with climate-only pathways: VSSI response by cluster, 1500–1900

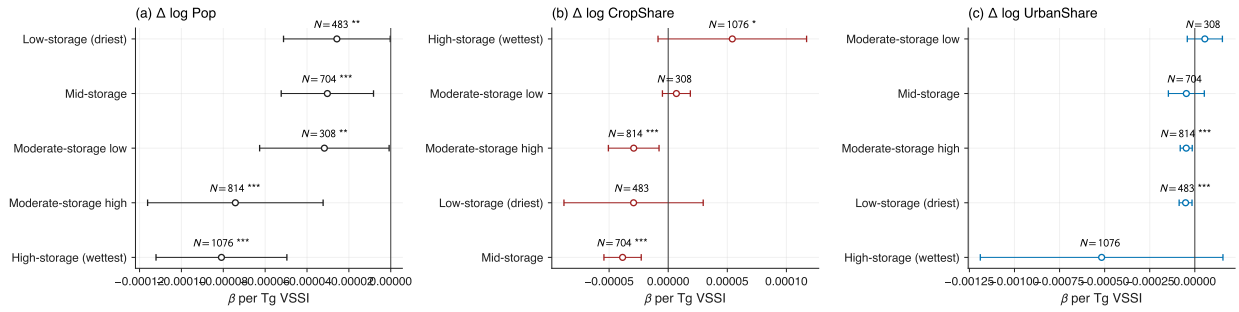


Figure 23: Climate-only pathway clustering: VSSI response by cluster across the three joint-VAR equations, 1500–1900. The asymmetry that the headline HYDE-clustered specification exhibits does not survive translation to climate-orthogonal pathway labels.

L.2 Wild-cluster bootstrap and label permutation

With twelve pathway-by-equation slopes in the headline joint VAR (four informative pathways times three equations), the asymptotic single-coefficient significance tests are vulnerable to multiple-testing inflation. We re-test the twelve slopes two ways.

First, a Webb (2014) six-point wild-cluster bootstrap (1,999 replications, restricted residuals re-signed at the country level) gives bootstrap p-values that track the cluster-robust asymptotic p-values closely (the two columns of Table 26 agree to within rounding). The cluster SEs are not catastrophically off; the asymptotic inference is well-calibrated. Under Bonferroni correction at the 5% level ($0.05/12 = 0.0042$), however, only two of the twelve slopes survive: the crop-dominant late and pastoral/mixed late demographic-margin coefficients (Table 26). The high-density intensive cropland coefficient ($p_{\text{boot}} = 0.017$) and the early-extensifier urban coefficient ($p_{\text{boot}} = 0.035$) fall short of Bonferroni significance, as do all remaining off-margin coefficients. A Benjamini-Hochberg FDR procedure at the 5% level returns the same two demographic-margin coefficients.

Table 26: Wild-cluster bootstrap p-values for the joint VAR pathway slopes, 1,999 Webb six-point replications. Bonferroni threshold: $p < 0.0042$. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Pathway	Equation	$\hat{\beta}$	p_{asyp}	p_{boot}	Bonferroni
Crop-dominant late	$\Delta \log P$	$-9.65 \cdot 10^{-5}$	$< 10^{-6}$ ***	0.000 ***	✓
Crop-dominant late	$\Delta \log s$	$+2.96 \cdot 10^{-5}$	0.258	0.301	
Crop-dominant late	$\Delta \log u$	$-5.16 \cdot 10^{-4}$	0.126	0.045 **	
Pastoral/mixed late	$\Delta \log P$	$-5.55 \cdot 10^{-5}$	$< 10^{-5}$ ***	0.000 ***	✓
Pastoral/mixed late	$\Delta \log s$	$+0.13 \cdot 10^{-5}$	0.913	0.931	
Pastoral/mixed late	$\Delta \log u$	$-6.02 \cdot 10^{-5}$	0.086 *	0.280	
High-density intensive	$\Delta \log P$	$-2.02 \cdot 10^{-5}$	0.215	0.233	
High-density intensive	$\Delta \log s$	$-3.73 \cdot 10^{-5}$	0.021 **	0.017 **	
High-density intensive	$\Delta \log u$	$+1.84 \cdot 10^{-5}$	0.751	0.823	
Early extensifiers	$\Delta \log P$	$-1.43 \cdot 10^{-5}$	0.121	0.146	
Early extensifiers	$\Delta \log s$	$-1.71 \cdot 10^{-5}$	0.186	0.210	
Early extensifiers	$\Delta \log u$	$-7.82 \cdot 10^{-5}$	0.040 **	0.035 **	

Second, a permutation test that re-shuffles the pathway labels across countries 300 times—preserving panel structure and within-country dynamics but breaking the link between cluster identity and country—asks how often a random labeling produces at least the observed number of significant coefficients per equation. For the demographic equation the answer is essentially always (median 4 significant slopes under permutation vs. 2 observed); the labeling actually reduces the count of significant demographic slopes relative to random. For the cropland equation, 48% of permutations produce at least one significant cropland slope. For the urban equation, 73% of permutations produce at least one significant urban slope. The observed cross-pathway pattern of one significant cropland and one significant urban response across four pathways is not distinguishable from what random labels produce.

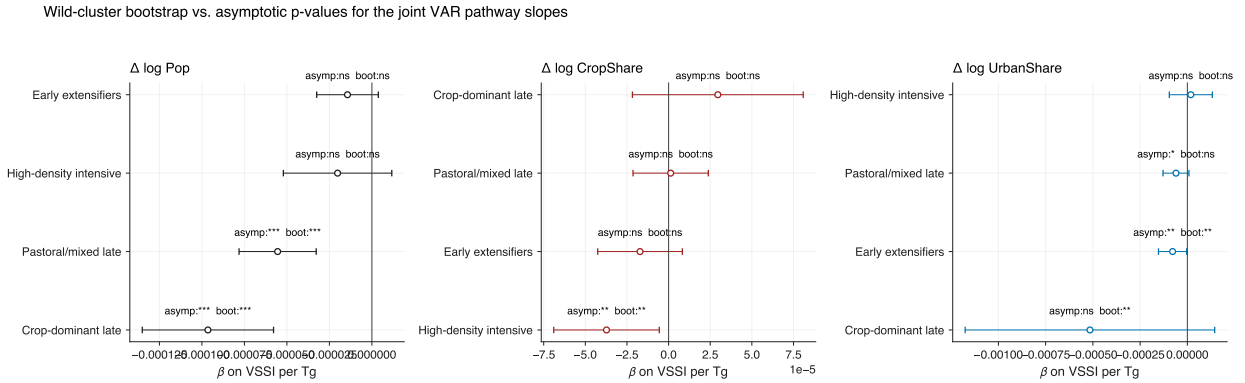


Figure 24: Joint VAR pathway slopes with both asymptotic and wild-cluster bootstrap significance markers. The bootstrap inference confirms the asymptotic inference; multiple-testing correction is what removes ten of the twelve slopes from significance.

L.3 Post-1700 sub-sample

HYDE 3.5 uses 100-year intervals before 1700 and decadal intervals afterwards; the 1500 and 1600 cells in the joint VAR panel are century-aggregated back-projections from a sparser historical source base than the post-1700 decadal cells. HYDE's pre-1850 demographic back-projection also uses

cropland and pasture extent as input variables, so the joint cropland-and-population dynamics in the pre-1700 portion of the panel are partly mechanical. We restrict to year ≥ 1700 (drops 305 century cells, $\sim 9\%$ of the panel) and re-run.

The post-1700 restriction differentially weakens the pathway coefficients. Table 27 reports the slopes across four sub-samples: full 1500–1900, post-1700, post-1750, and post-1800. The crop-dominant late demographic margin survives every cut ($p \leq 0.018$). The pastoral/mixed demographic margin collapses to $p = 0.95$ post-1700: that significance came entirely from the century-resolution cells. The high-density intensive cropland margin weakens to $p = 0.076$ post-1700 and to $p = 0.26$ post-1750. The early-extensifier urban margin weakens to $p = 0.28$ post-1700.

Table 27: Joint VAR pathway slopes across temporal sub-samples. Coefficients in units of 10^{-5} per Tg VSSI. Stars: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Pathway	Sub-sample	$\hat{\beta}^{\text{Pop}} (p)$	$\hat{\beta}^{\text{Crop}} (p)$	$\hat{\beta}^{\text{Urb}} (p)$
Crop-dominant late	1500–1900 (full)	−9.65 ***	+2.96	−51.6
	1700–1900	−4.10 **	+36.3 **	+9.95
	1750–1900	−5.66 ***	+39.1 **	+0.19
	1800–1900	−11.9 **	+78.4 ***	−0.13
Pastoral/mixed late	1500–1900 (full)	−5.55 ***	+0.13	−6.02 *
	1700–1900	−0.11	−4.79	−0.61
	1750–1900	−0.48	−5.99	+0.79
	1800–1900	−2.22	−10.0	−2.45
High-density intensive	1500–1900 (full)	−2.02	−3.73 **	+1.84
	1700–1900	−0.43	−4.16 *	−8.44
	1750–1900	−0.23	−3.32	−9.18
	1800–1900	−5.94 *	−2.26	−4.81
Early extensifiers	1500–1900 (full)	−1.43	−1.71	−7.82 **
	1700–1900	−3.69 *	+10.7	−3.13
	1750–1900	−2.93	+15.7 *	−2.82 *
	1800–1900	−6.11	+20.7	−3.38

The crop-dominant late demographic margin is the only coefficient that retains significance across all four sub-samples. The other three margins of the original three-margin asymmetry either weaken substantially (high-density intensive cropland, early extensifier urban) or vanish (pastoral/mixed demographic) when the century-resolution HYDE cells are dropped.

Joint VAR pathway coefficients: full panel vs. post-1700 (drop century cells)

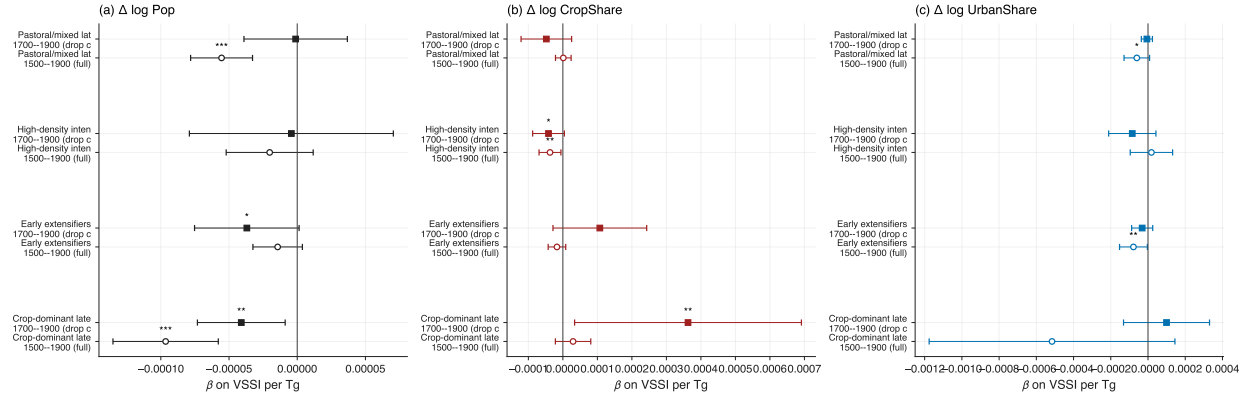


Figure 25: Joint VAR pathway coefficients across temporal sub-samples. Full panel (open circles) versus post-1700 decadal cells only (filled squares). Significance markers are nominal asymptotic.

L.4 Reading the robustness exercises

The three exercises target related but distinct issues with the joint-VAR cross-pathway asymmetry: pathway endogeneity (L.1), multiple-testing inflation (L.2), and HYDE temporal-imputation artifacts (L.3). Each exercise weakens a different component of the original headline. Taking the three together, what is statistically defensible is a clean demographic-margin response to volcanic forcing that is concentrated in crop-dominant late countries and is robust across pathway typology, multiple-testing correction, and HYDE sub-sample. What is not statistically defensible at conventional levels is the further claim that intensive and early-extensifier pathways adjust on non-demographic margins: those slopes are individually nominally significant in the full HYDE-clustered specification but do not survive any one of the three robustness exercises.

A noteworthy and previously unreported result emerges from the post-1700 restriction: the crop-dominant late pathway shows a sharply *positive* cropland-share response to volcanic forcing once the century-resolution cells are dropped ($\hat{\beta} = +3.6 \cdot 10^{-4}$ per Tg post-1700, $p = 0.031$; $+7.8 \cdot 10^{-4}$, $p = 0.003$ in 1800–1900). After volcanic cooling, crop-dominant late economies expanded their cropland share—a Boserupian intensification response in the decade following a cold shock. This finding is robust to the temporal sub-sample but is masked in the full 1500–1900 panel by the offsetting pre-1700 century cells. We note it here as a candidate result for follow-up work.

The honest reading of the joint VAR is therefore: it identifies a clean demographic effect within a multi-margin framework, surfaces an unexpected Boserupian cropland response in the modern-era crop-dominant subsample, and motivates future work with finer temporal demographic resolution that can resolve the within-decade dynamics of the cropland and urban margins.

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